## Closed Mapping

**Def:**
A map

$$f : (X, \mathcal{T}_X) \to (Y, \mathcal{T}_Y)$$

is called a closed map iff the image of each closed subset of $X$ is closed in $Y$.

## Closed Mapping

**Def:**
A map

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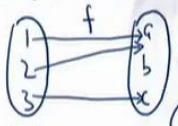
is called a closed map iff the image of each closed subset of $X$ is closed in $Y$.

$\forall\ U \underset{\text{closed}}{\subset} X \Rightarrow f(U) \underset{\text{closed}}{\subset} Y$

## Closed Mapping

Ex: $X = \{1,2,3\}$, $\mathcal{T}_X = \{\emptyset, \{1\}, \{1,2\}, X\}$

$Y = \{a,b,c\}$, $\mathcal{T}_Y = \{\emptyset, \{a,b\}, \{b\}, Y\}$




$\mathcal{C}_X = \{X, \{2,3\}, \{3\}, \emptyset\}$

$\mathcal{C}_Y = \{Y, \{c\}, \{a,c\}, \emptyset\}$

$f(\emptyset) = \emptyset$ ✓

$f(\{3\}) = \{c\}$ ✓ | $f(\{2,3\}) = \{a,c\}$ ✓ | $f(X) = \{a,c\}$ ✓

## Topology

Projection Mapping

## Projection Mapping

Projection mapping is an example of mapping that is not closed in general.

But let us see first "What is projection map?"

## Projection Mapping

**Projection Map:**
Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be two topological spaces. The maps

Product top:

$$\begin{aligned} \pi_1 : (X \times Y, \mathcal{T}_{X\times Y}) &\to (X, \mathcal{T}_X) \\ (x,y) &\mapsto x \end{aligned}$$

$$\begin{aligned} \pi_2 : (X \times Y, \mathcal{T}_{X\times Y}) &\to (Y, \mathcal{T}_Y) \\ (x,y) &\mapsto y \end{aligned}$$

are called projection maps, where $\mathcal{T}_{X\times Y}$ is product topology on $X \times Y$. $\pi_1$ is first projection and $\pi_2$ is second projection.

## Projection Mapping

**Fact:**
Projection map may or may not be closed.
**Justification:**
Consider the first projection

$$\pi_1 : (\mathbb{R} \times \mathbb{R}, \mathcal{T}_{\mathbb{R}\times\mathbb{R}}) \to (\mathbb{R}, \mathcal{T}_u)$$

The set $A = \{(x,y) |\ y = \tan x\ ,\ \frac{-\pi}{2} < x < \frac{\pi}{2}\}$ is closed in $\mathbb{R} \times \mathbb{R}$ but

$$\pi_1(A) = \left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$$

## Projection Mapping

**Fact:**
Projection map may or may not be closed.
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The set $A = \{(x,y) |\ y = \tan x\ ,\ \frac{-\pi}{2} < x < \frac{\pi}{2}\}$ is closed in $\mathbb{R} \times \mathbb{R}$ but

$$\pi_1(A) = \left(\frac{-\pi}{2}, \frac{\pi}{2}\right) \subset \mathbb{R}.$$

Not closed

## Topology

Continuity and Open Mapping

## Continuity and Open Mapping

We have studied

- Continuous Mapping
- Open Mapping
- Closed Mapping

**Questions:**

1. Is a continuous map an open map?
2. Is an open map a continuous map?

## Continuity and Open Mapping

**Example 1:**
Let $X = \{a, b, c, d\}$ with $\mathcal{T}_X = \{\emptyset, \{b, c\}, \{a, b, c\}, X\}$
and
$Y = \{1, 2, 3\}$ with $\mathcal{T}_Y = \{\emptyset, \{1\}, \{1, 2\}, Y\}$.

Consider the following map

$$f : X \to Y$$

defined as $f(x) = 2$ for all $x \in X$.

## Continuity and Open Mapping

**Example 1:**
Let $X = \{a, b, c, d\}$ with $\mathcal{T}_X = \{\emptyset, \{b, c\}, \{a, b, c\}, X\}$
and
$Y = \{1, 2, 3\}$ with $\mathcal{T}_Y = \{\emptyset, \{1\}, \{1, 2\}, Y\}$.

Consider the following map

$$f : X \to Y$$

Continuous.

$f(\{b,c\}) = \{2\}$ ∩ Not open in $Y$

defined as $f(x) = 2$ for all $x \in X$.

## Continuity and Open Mapping

**Example 2:**
Let $X = \mathbb{R}$ with usual topology and $Y = \mathbb{R}$ with lower limit topology.

Consider the identity map

$$f : X \to Y$$
$$x \mapsto x$$

$[a, b) \subset \mathbb{R}_l$

$f^{-1}[a,b) = [a,b) \subset \mathbb{R}_u$

$f$ is not cont.

$(a,b) \in \mathcal{B}_u \quad f((a,b)) = (a,b) \in \mathcal{T}_l$

## Continuity and Open Mapping

**Answers:**

1. Continuous map may or may not be open map.
2. Open map may or may not be continuous map.

## Topology

Openness and Restriction Mapping

## Openness and Restriction...

Recall the restriction mapping and the fact about the continuity of restriction mapping.
Now the question is

"Can we prove the same kind of result for restriction of an open mapping?".

## Openness and Restriction...

**Example:** Let $X = \{a, b, c, d\}$ with $\mathcal{T}_X = \{\emptyset, \{a\}, \{a, b\}, \{a, b, c\}, X\}$ and $Y = \{1, 2, 3\}$ with $\mathcal{T}_Y = \{\emptyset, \{1\}, \{1, 2\}, Y\}$.

Consider the map $f : X \to Y$ defined as

$$f(a) = 1,\ f(b) = 2,\ f(c) = 3,\ f(d) = 2$$

Let $A = \{b, c, d\} \subset X$ with relative topology $\mathcal{T}_A = \{\emptyset, \{b\}, \{b, c\}, A\}$. Now the restriction map $f_A : A \to Y$ is defined as

$$f_A(b) = 2,\ f_A(c) = 3,\ f_A(d) = 2$$

## Openness and Restriction...

Now $f$ is an open map but $f_A$ is not an open map. Because $\{b\}$ is an open subset of $A$ but

$$f_A(\{b\}) = \{2\} \underset{\text{NotOpen}}{\subset} Y$$

**Remark:**

Restriction of an open map may or may not be an open map.

KHALIL SAHIR
MC200400708

## Topology

Homeomorphism and Homeomorphic Spaces

## Homeomorphism...

- $f$ is one-one.
- $f$ is onto

**Homeomorphism:**
A map $f : X \to Y$ is called a homeomorphism between two topological spaces $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ iff

1. $f$ is bijective.
2. $f$ is continuous.
3. $f^{-1} : Y \to X$ is continuous.

## Homeomorphism...

$\forall\ U \underset{\text{open}}{\subset} Y$

$f^{-1}(U) \underset{\text{open}}{\subset} X$

**Homeomorphism:**
A map $f : X \to Y$ is called a homeomorphism between two topological spaces $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ iff

1. $f$ is bijective.
2. $f$ is continuous.
3. $f^{-1} : Y \to X$ is continuous.

## Homeomorphism...

$\forall\ V \underset{\text{open}}{\subset} X$

$\left(f^{-1}\right)^{-1}(V) \underset{\text{open}}{\subset} Y$

**Homeomorphism:**
A map $f : X \to Y$ is called a homeomorphism between two topological spaces $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ iff

1. $f$ is bijective.
2. $f$ is continuous.
3. $f^{-1} : Y \to X$ is continuous.

## Homeomorphism...

**Homeomorphic Spaces:**
Two topological spaces $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ are called homeomorphic iff there exists a homeomorphism $f : X \to Y$. We also say such spaces topologically equivalent spaces.

$X \simeq Y$

## Topology

Examples:
Homeomorphism

## Examples: Homeomorphism

Let us see some examples of homeomorphism and homeomorphic spaces.

## Examples: Homeomorphism

**Example 1:** Consider
$X = \{a, b, c, d\}$ with $T_X = \{\emptyset, \{a, b\}, \{c, d\}, X\}$
$Y = \{1, 2, 3, 4\}$ with $T_Y = \{\emptyset, \{2, 3\}, \{1, 4\}, Y\}$
and the map $f : X \to Y$ defined as

$$f(a) = 2,\ f(b) = 3,\ f(c) = 4,\ f(d) = 1$$

$f^{-1}(\{2,3\}) = \{a,b\} \subseteq_{open} X$ | $f^{-1}(Y) = X \subseteq_{open} X$.
$f^{-1}(\{1,4\}) = \{c,d\} \subseteq_{open} X$

## Examples: Homeomorphism

**Example 1:** Consider
$X = \{a, b, c, d\}$ with $T_X = \{\emptyset, \{a, b\}, \{c, d\}, X\}$
$Y = \{1, 2, 3, 4\}$ with $T_Y = \{\emptyset, \{2, 3\}, \{1, 4\}, Y\}$
and the map $f : X \to Y$ defined as

$$f(a) = 2,\ f(b) = 3,\ f(c) = 4,\ f(d) = 1$$

Now $f^{-1} : Y \to X$
$f^{-1}(1) = d$, $f^{-1}(2) = a$, $f^{-1}(3) = b$, $f^{-1}(4) = c$
bij. + Cont + $f^{-1}$ Cont.
$f^{-1}$ is Cont.

## Examples: Homeomorphism

**Example 2:** Consider
$X = [0, 1]$, $Y = [0, 2]$ as subspaces of $(\mathbb{R}, \mathcal{T}_u)$
and the map $f : X \to Y$ defined as

$$f(x) = 2x \quad \forall x \in X$$

- $f$ is bijection.
- $f$ is Cont.
- $f^{-1} : Y \to X$ $\quad f^{-1}(y) = \frac{y}{2}$
  So $f^{-1}$ is Cont.

$X = [0,1]$ ⫋ $Y = [0,2]$

## Examples: Homeomorphism

**Example 3:** Consider
$X = (\frac{-\pi}{2}, \frac{\pi}{2})$ as subspace of $(\mathbb{R}, \mathcal{T}_u)$ and $Y = \mathbb{R}$ with usual topology and the map $f : X \to Y$ defined as

$$f(x) = \tan x \quad \forall x \in X$$

- $f$ is bij. ✓
- $f$ is Cont. ✓
- $f^{-1}$ is Cont ✓

$X = (\frac{-\pi}{2}, \frac{\pi}{2})$ ⫋ $Y = \mathbb{R}$

## Topology

**Bicontinuity and Open Mapping**

## Bicontinuity and Open Mapping

$f : X \to Y$
$\forall\ U \subseteq_{open} Y$, $f^{-1}(U) \subseteq_{open} X$

$f^{-1} : Y \to X$
$\forall\ V \subseteq_{open} X$, $(f^{-1})^{-1}(V) = f(V) \subseteq_{open} Y$
open mapping

Recall homeomorphism.
A bijective map $f : X \to Y$ such that $f$ and $f^{-1}$ both continuous or in other words $f$ is bicontinuous.

## Bicontinuity and Open Mapping

A map $f : X \to Y$ is a homeomorphism between two topological spaces $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ iff

1. $f$ is bijective.
2. $f$ is continuous.
3. $f$ is open map.

## Topology

Bicontinuity and Bijection

## Bicontinuity and Bijection

**Question:**
Is there any relation between bicontinuity and bijection?

More specifically.

- Does bijection imply bicontinuity?
- Does bicontinuity imply bijection?

## Bicontinuity and Bijection

**Example 1:** Consider
$X = \{a, b, c, d\}$ with $T_X = \{\emptyset, \{a\}, \{c, d\}, \{a, c, d\}, X\}$
$Y = \{1, 2, 3, 4\}$ with $T_Y = \{\emptyset, \{1\}, \{2\}, \{1, 2\}, Y\}$
and the map $f : X \to Y$ defined as

$$f(a) = 1,\ f(b) = 2,\ f(c) = 3,\ f(d) = 4$$

## Bicontinuity and Bijection

**Example 1:** Consider
$X = \{a, b, c, d\}$ with $T_X = \{\emptyset, \{a\}, \{c, d\}, \{a, c, d\}, X\}$
$Y = \{1, 2, 3, 4\}$ with $T_Y = \{\emptyset, \{1\}, \{2\}, \boxed{\{1, 2\}}, Y\}$
and the map $f : X \to Y$ defined as

$$f(a) = 1,\ f(b) = 2,\ f(c) = 3,\ f(d) = 4$$

$f$ is bijective. but $f$ is not cont.

$f^{-1}(\{1,2\}) = \{a, b\} \subset X$ (Not open)

Bijection $\not\Rightarrow$ Bicont.

## Bicontinuity and Bijection

**Example 2:** Consider
$X = \{a, b, c, d\}$ with $T_X = \{\emptyset, X\}$
$Y = \{1, 2, 3, 4\}$ with $T_Y = \{\emptyset, \{1\}, Y\}$
and the map $f : X \to Y$ defined as

$$f(a) = 1,\ f(b) = 1,\ f(c) = 1,\ f(d) = 1$$

## Bicontinuity and Bijection

**Example 2:** Consider
$X = \{a, b, c, d\}$ with $T_X = \{\emptyset, X\}$
$Y = \{1, 2, 3, 4\}$ with $T_Y = \{\emptyset, \{1\}, Y\}$
and the map $f : X \to Y$ defined as

$$f(a) = 1,\ f(b) = 1,\ f(c) = 1,\ f(d) = 1$$

$f$ is (cont + open) $\not\Rightarrow$ Bijective.

## Topology

Homeomorphism as Equivalence Relation

## Homeomorphism...

**Equivalence Relation:**
A binary relation $\sim$ on a set $X$ is said to be an equivalence relation iff $\sim$ satisfies the following three properties for all $a, b, c \in X$

1. Reflexive: $a \sim a$
2. Symmetric: $a \sim b$ iff $b \sim a$
3. Transitive: If $a \sim b$ and $b \sim c$ then $a \sim c$

## Homeomorphism...

**Proposition:**
Homeomorphism of topological spaces is an equivalence relation.

**Proof:** • $X \approx X$ $\quad Id: X \to X$, $x \mapsto x$

• Let $X \approx Y$ $\quad f: X \to Y$ Hom. Bijectn + Bicont.

$f^{-1}: Y \to X$ Bijectn + Cont + $f$ is cont

$\Rightarrow X \approx Y \Leftrightarrow Y \approx X$.

## Homeomorphism...

• $X \approx Y$ , $Y \approx Z$

$f: X \to Y$ Bijectn + Bicont. | $g: Y \to Z$ Bijective + Bicont.

$g \circ f: X \to Z$

Bijective. + $g \circ f$ is bicont.

$\to g \circ f$ is cont

$f^{-1} \circ g^{-1}$ is cont $\Rightarrow X \approx Z$.

## Topology

Topological Properties

## Topological Properties

**Topological Property:**
A property "$P$" of a topological space $(X, \mathcal{T})$ is called "Topological Property" if every space homeomorphic to $X$ has property $P$.
i.e.
A property that is unchanged under homeomorphism.
Topological property is also called topological invariant.

## Topological Properties

**Some Topological Properties:**
- Cardinality of set $X$.
- Cardinality of $\mathcal{T}$.
- Connectedness of space $X$.
- Being Discrete.

## Topological Properties

$X = [0, 2]$ , $Y = [0, 4]$

$L_X = 2$ , $L_Y = 4$

$f: X \to Y$

$x \mapsto 2x$ Biject. + Cont + Inv. of $f$ is also cont

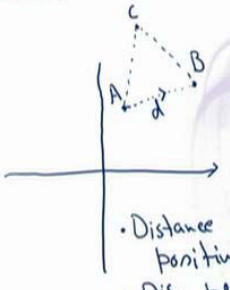
$X \approx Y$

**Some Non Topological Properties:**
- Length
- Area
- Boundedness

## Topology

Metric on a Set

## Metric on a Set




A metric is a distance function i.e.
A function that defines a distance between each pair of elements of a set.

- Distance between A and B is always positive.
- Dis. bet. A and A is "0"
- "0" is the dis. only from A to A.
- Dis bet. A and B = Dis bet. B and A.
- dis. bet A and B $\leq$ dis. bet A and C + dis bet C and B.

## Metric on a Set

**Metric:**
A metric on a non empty set $X$ is a real valued function $d$ defines on $X \times X$, i.e

$$d : X \times X \longrightarrow \mathbb{R}$$

such that for all $x, y, z \in X$, the following conditions are satisfied:

1. $d(x, y) \geq 0$, equality holds iff $x = y$
2. $d(x, y) = d(y, x)$ (Symmetry)
3. $d(x, z) \leq d(x, y) + d(y, z)$ (Triangle Inequality)

## Metric on a Set

**Metric Space:**
A set $X$ with a metric $d$ defined on it is called a metric space.

## Topology

Examples of Metric

## Examples of Metric

Let us see some examples of metrics and verify all the three axioms for each metric.

**Example 1:**
Let $X = \mathbb{R}$ and $d$ is defined as $d(x, y) = |x - y|$ where $x, y \in \mathbb{R}$.

## Examples of Metric

- $d(x,y) = |x-y| \geq 0$

  $|x-y| = 0 \iff x - y = 0$

  $x = y$

- $d(x,y) = |x-y| = |-(y-x)|$

  $= |y-x| = d(y,x)$

- $x, y, z \in \mathbb{R}$

  $d(x,z) = |x-z| = |x-y+y-z| \leq \underbrace{|x-y|}_{d(x,y)} + \underbrace{|y-z|}_{d(y,z)}$

  $d(x,z) \leq d(x,y) + d(y,z)$

$|a| = \begin{cases} a & \text{if } a \geq 0 \\ -a & \text{if } a < 0 \end{cases}$

$|x| = |-x|$

$|a+b| \leq |a| + |b|$

## Examples of Metric

**Example 2:**
Let $X = C[0, 1]$ (the set of all real valued continuous functions on $[0, 1]$) and $d$ is defined as

$$d(f, g) = \int_0^1 |f(x) - g(x)|dx$$

where $f, g \in C[0, 1]$.

## Examples of Metric

**Example 2:**
Let $X = C[0, 1]$ (the set of all real valued continuous functions on $[0, 1]$) and $d$ is defined as

$$d(f, g) = \int_0^1 |f(x) - g(x)|dx$$

where $f, g \in C[0, 1]$.

Sol: $d(f,g) = \int_0^1 |f(x) - g(x)|dx \geq 0$

$d(f,g) = 0 \iff \int_0^1 |f(x) - g(x)|dx = 0$

$\iff f(x) - g(x) = 0 \iff f(x) = g(x) \quad \forall x \in [0,1]$

## Examples of Metric

- $d(f,g) = \int_0^1 |f(x) - g(x)|dx = \int_0^1 |g(x) - f(x)|dx$

  $= d(g,f)$

- $d(f,h) = \int_0^1 |f(x) - h(x)|dx = \int_0^1 |f(x) - g(x) + g(x) - h(x)|dx$

  $\leq \int_0^1 |f(x) - g(x)|dx + \int_0^1 |g(x) - h(x)|dx$

  $= d(f,g) + d(g,h)$. □

## Topology

More Examples of Metric

## More Examples of Metric

Let us see some more examples of metrics.

**Example 3:**
Let $X$ be any nonempty set and $d$ is defined as

$$d(x,y) = \begin{cases} 0, & \text{if } x = y \\ 1, & \text{if } x \neq y \end{cases}$$

where $x, y \in \mathbb{R}$.

## More Examples of Metric

- $d(x,y) = \begin{cases} 0 \\ 1 \end{cases} \geq 0$

  $d(x,y) = 0 \iff x = y$

- $d(x,y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases} = \begin{cases} 0 & \text{if } y = x \\ 1 & \text{if } y \neq x \end{cases}$

  $= d(y,x)$

- $x, y, z \in X \qquad d(x,z) = 1 < 1+1 = d(x,y) + d(y,z)$

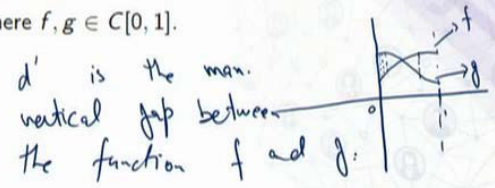
## More Examples of Metric

**Example 4:**
Let $X = C[0,1]$ (the set of all real valued continuous functions on $[0,1]$) and $d'$ is defined as

$$d'(f,g) = sup\{|f(x) - g(x)| \ : \ x \in [0,1]\}$$

where $f, g \in C[0,1]$.

$d'$ is the max. vertical gap between the function $f$ and $g$:




## Topology

Usual Metric on $\mathbb{R}^n$

## Usual Metric on $\mathbb{R}^n$

Recall the metric defined on $\mathbb{R}$ in "Example 1."

$$d(x,y) = |x - y|$$

where $x, y \in \mathbb{R}$.
This is the usual metric in $\mathbb{R}$.

We can also write it in the following way

$$d(x,y) = \sqrt{(x-y)^2}$$

## Usual Metric on $\mathbb{R}^n$

**Usual Metric on $\mathbb{R}^2$**
Consider $\mathbb{R}^2$ and $d$ is defined as

$$d(x,y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

where $x = (x_1, x_2), y = (y_1, y_2) \in \mathbb{R}^2$.

## Usual Metric on $\mathbb{R}^n$

**Usual Metric on $\mathbb{R}^n$**
Consider $\mathbb{R}^n$ and $d$ is defined as

$$d(x,y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \cdots + (x_n - y_n)^2}$$

where $x = (x_1, x_2, \cdots, x_n), y = (y_1, y_2, \cdots, y_n) \in \mathbb{R}^n$.

## Topology

Unusual Metrics on $\mathbb{R}^n$

## Unusual Metrics on $\mathbb{R}^n$

Let us see some unusual metrics defined on $\mathbb{R}^2$. These definitions can be generalized for $\mathbb{R}^n$ with $n > 2$.

## Unusual Metrics on $\mathbb{R}^n$

**Example 1:**
Consider $\mathbb{R}^2$ and $d_1$ is defined as

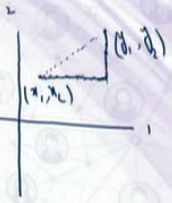
$$d_1(x, y) = max(|x_1 - y_1|, |x_2 - y_2|)$$

where $x = (x_1, x_2), y = (y_1, y_2) \in \mathbb{R}^2$.

$x = (2, 3)$, $y = (7, 15)$

$d_1(x, y) = \max(|2-7|, |3-15|)$

$= \max(5, 12) = 12$




## Unusual Metrics on $\mathbb{R}^n$

**Example 2:**
Consider $\mathbb{R}^2$ and $d_2$ is defined as

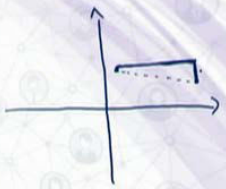
$$d_2(x, y) = |x_1 - y_1| + |x_2 - y_2|$$

where $x = (x_1, x_2), y = (y_1, y_2) \in \mathbb{R}^2$.

$x = (1, 3)$ $y = (7, 2)$

$d_2(x, y) = |1-7| + |3-2|$

$= 6 + 1 = 7$



## Topology

Open Ball

## Open Ball

**Def:**
Consider $X$ with a metric $d$. For any $x \in X$ and any real number $r > 0$, the set of all points $y \in X$ whose distance from $x$ is less than $r$ is called the **open ball** of radius $r$ centered at $x$.
i.e.

$$B(x, r) = \{y \in X | d(x, y) < r\}$$

## Open Ball

**Example:**
Recall $\mathbb{R}$ with usual metric $d = |x - y|$. Consider $x = 2$ and $r = 1$.
Let us see what is $B(2, 1)$?

## Open Ball

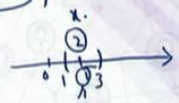
**Example:**
Recall $\mathbb{R}$ with usual metric $d(x,y) = |x - y|$. Consider $x = 2$ and $r = 1$.
Let us see what is $B(2, 1)$?

$B(2,1) = \{y \in \mathbb{R} \mid d(2, y) = |2 - y| < 1\} = (1, 3)$

$|2 - y| < 1$

$\Rightarrow -1 < 2 - y < 1$

$\Rightarrow 1 < y < 3$



# Topology

Examples: Open Ball

# Examples: Open Ball

We have already discussed one simple example of open ball. Now we will discuss some more examples of open balls with respect to different metrics.

# Examples: Open Ball

**Example 1:**
Recall $\mathbb{R}^2$ with usual metric
$d(x,y) = \sqrt{(x_1-y_1)^2+(x_2-y_2)^2}$ where
$x=(x_1,x_2), y=(y_1,y_2) \in \mathbb{R}^2$.

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$B(x,1) = \{y \in \mathbb{R}^2 \mid d(x,y) < 1\}$

$x=(0,0)$

$r=1$

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$B(x,r) = \{y \in \mathbb{R}^2 \mid d(x,y) < r\}$ $\quad r>0$

# Examples: Open Ball

**Example 2:**
Recall trivial metric on a set $X$. i.e.

$$d(x,y) = \begin{cases} 0 & \text{if } x=y \\ 1 & \text{if } x \neq y \end{cases}$$

where $x, y \in X$.

# Examples: Open Ball

**Example 2:**
Recall trivial metric on a set $X$. i.e.

$$d(x,y) = \begin{cases} 0 & \text{if } x=y \\ 1 & \text{if } x \neq y \end{cases}$$

where $x, y \in X$.

$B(x,r) = \{y \in X \mid d(x,y) < r\} = \{x\}$ $\quad r \leq 1$

$B(x,r) = \{y \in X \mid d(x,y) < r\} = X.$ $\quad r > 1$

# Examples: Open Ball

**Example 3:**
Recall $\mathbb{R}^2$ with $d_1(x,y) = max(|x_1-y_1|, |x_2-y_2|)$ where
$x=(x_1,x_2), y=(y_1,y_2) \in \mathbb{R}^2$.

$x=(0,0)$

$B(x,1)$

## Examples: Open Ball

**Example 4:**
Recall $C[0,1]$ with $d(f,g) = \int_0^1 |f(x) - g(x)|dx$ where $f, g \in C[0,1]$.

$B(f, \lambda)$

$\lambda = 3$

## Topology

Open Ball as Open Set



## Open Ball as Open Set

**Lemma:**
Let $d$ be a metric defined on a nonempty set $X$. For $x \in X$ and a positive real

## Open Ball as Open Set

Proof: Let $y \in B(x, \lambda)$

$\Rightarrow \quad d(x,y) < \lambda$

$\Rightarrow \quad \lambda - d(x,y) > 0$

Let $\lambda' := \lambda - d(x,y)$

Now we want to show that

$B'(y, \lambda') \subset B(x, \lambda)$

## Topology

Open Balls Centered at One Point



## Open Balls Centered at One Point

**Fact:**
Let $B_1(x, r_1)$ and $B_2(x, r_2)$ be two open balls centered at $x$ with radius $r_1$ and $r_2$ respectively. Then either $B_1(x, r_1) \subset B_2(x, r_2)$ or $B_2(x, r_2) \subset B_1(x, r_1)$.



## Open Balls Centered at One Point

Proof: $B_1(x, \lambda_1)$, $B_2(x, \lambda_2)$

$\lambda_1, \lambda_2 \in \mathbb{R}^+$

$\Rightarrow \quad \lambda_1 \leq \lambda_2$ or $\lambda_2 \leq \lambda_1$

$\Rightarrow \quad B_1(x, \lambda_1) \subset B_2(x, \lambda_2)$ | $B_2(x, \lambda_2) \subset B_1(x, \lambda_1)$

$y \mid d(x,y) < \lambda_1$

## Topology

Properties of
Collection of Open Balls

## Properties of Collection...

**Theorem:**
Consider $X$ with a metric $d$. Let $\mathcal{B}$ be the collection of all open balls in $X$. Then $\mathcal{B}$ satisfies the following properties.

1. For all $x \in X$ there exists $B(x,r) \in \mathcal{B}$
2. For all $z \in B_1(x,r_1) \cap B_2(y,r_2)$ there exists $B(z,r)$ such that $z \in B(z,r) \subset B_1(x,r_1) \cap B_2(y,r_2)$

## Properties of Collection...

Proof. $X$, $d$, $\mathcal{B}$

$x \in X \quad \lambda > 0$

$x \in B(x,\lambda) \in \mathcal{B}$

② → Let $z \in B_1(x,\lambda_1) \cap B_2(y,\lambda_2)$

$\Rightarrow z \in B_1(x,\lambda_1)$ and $z \in B_2(y,\lambda_2)$

$\Rightarrow \exists\ B_3(z,\lambda_3) \subset B_1(x,\lambda_1)$ $\Rightarrow \exists\ B_4(z,\lambda_4) \subset B_2(y,\lambda_2)$

$\lambda_3, \lambda_4 \in \mathbb{R}^+$

$\lambda_3 < \lambda_4$

## Properties of Collection...

$B_3(z,\lambda_3) \subset B_4(z,\lambda_4) \subset B_2(y,\lambda_2)$

$z \in B_3(z,\lambda_3) \subset B_1(x,\lambda_1) \cap B_2(y,\lambda_2)$

$B_3(z,\lambda_3) = B(z,\lambda)$

□

## Topology

Metric Topology

## Metric Topology

For a non empty set $X$ we defined a metric $d$.
With respect to this metric we defined open balls.
And considered $\mathcal{B}$ "Collection of all open balls in $X$."
This collection $\mathcal{B}$ satisfies the properties of "Basis."
So $\mathcal{B}$ is able to generate a topology on $X$.

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## Metric Topology

**Metric Topology:**
Let $X$ be a nonempty set with metric $d$. The topology $\mathcal{T}$ on $X$ generated by the set of all open balls in $X$ with respect to $d$ is called the "Metric Topology."
We also say that a topology $\mathcal{T}$ on $X$ induced by the metric $d$.

**Metric Space:**
Consider $X$ with a metric $d$, this $d$ induces a topology the "metric topology." $(X,d)$ is called a metric space.

## Topology

Examples: Metric
Topology

## Examples: Metric Topology

Let us see some examples of metric topologies. In these examples we will generate the topologies induced by already defined metrics.

## Examples: Metric Topology

**Example 1:**
Recall $\mathbb{R}$ with usual metric $d(x,y) = |x-y|$ where $x, y \in \mathbb{R}$.
Here $\mathcal{B} = \{B(x,r) | x \in X \quad r > 0\}$ is the set of all open intervals.

And we know that the topology generated by the set of all open intervals is usual topology.
So the metric topology induced by $d(x,y) = |x-y|$ on $\mathbb{R}$ is usual topology on $\mathbb{R}$.

## Examples: Metric Topology

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Recall $\mathbb{R}$ with usual metric $d(x,y) = |x-y|$ where $x, y \in \mathbb{R}$.
Here $\mathcal{B} = \{B(x,r) | x \in X \quad r > 0\}$ is the set of all open intervals.

$B(x,\lambda) = (x-\lambda, x+\lambda)$

And we know that the topology generated by the set of all open intervals is usual topology.
So the metric topology induced by $d(x,y) = |x-y|$ on $\mathbb{R}$ is usual topology on $\mathbb{R}$.

## Examples: Metric Topology

**Example 2:**
Recall $\mathbb{R}^2$ with usual metric
$d(x,y) = \sqrt{(x_1-y_1)^2 + (x_2-y_2)^2}$ where
$x = (x_1, x_2), y = (y_1, y_2) \in \mathbb{R}^2$.
Here $\mathcal{B} = \{B(x,r) | x \in X \quad r > 0\}$ is the set of all open disks.

And we know that the topology generated by the set of all open disks is usual topology on $\mathbb{R}^2$. So the metric topology induced by $d(x,y) = \sqrt{(x_1-y_1)^2 + (x_2-y_2)^2}$ on $\mathbb{R}^2$ is the usual topology on $\mathbb{R}^2$.

## Examples: Metric Topology

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## Examples: Metric Topology

**Example 3:**
Recall trivial metric on a set $X$. i.e.

$$d(x,y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}$$

where $x, y \in X$. Here
$\mathcal{B} = \{B(x,r) | x \in X \quad r > 0\} = \{\{x\}, X | x \in X\}$.

And we know that the topology generated by the set of all single tons is discrete topology. So the metric topology induced by the above metric is the discrete topology on $X$.

## Examples: Metric Topology

**Example 3:**
Recall trivial metric on a set $X$. i.e.

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$\mathcal{B} = \{B(x,r) | x \in X \quad r > 0\} = \{\{x\}, X | x \in X\}$.

And we know that the topology generated by the set of all single tons is discrete topology. So the metric topology induced by the above metric is the discrete topology on $X$.

## Topology

Metrizable Spaces

## Metrizable Spaces

Consider a non empty set $X$ with a metric $d$ defined on $X$. Now this metric induces a topology on $X$. Let's reverse this process, i.e.
We have a set $X$ with a topology $\mathcal{T}$, now the question is
"Dose there exist a metric $d$ such that $d$ on $X$ induces the topology $\mathcal{T}$?"
The answer depends on the selection of $\mathcal{T}$.

## Metrizable Spaces

**Metrizable Space:**
A topological space $(X, \mathcal{T})$ is called metrizable space if there exists a metric $d$ on $X$ that induces the topology $\mathcal{T}$.
**Remark:**
Not all spaces are metrizable.

## Topology

Examples: Metrizable Spaces

## Examples: Metrizable Spaces

Let us see some examples of metrizable spaces, i.e. the topological space such that there exists a metric that induces the given topology.

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## Examples: Metrizable Spaces

**Example 1:**
Consider $\mathbb{R}$ with usual topology. Recall usual metric $d(x,y) = |x - y|$ on $\mathbb{R}$.
Now the topology induced by $d(x,y) = |x - y|$ on $\mathbb{R}$ is the usual topology on $\mathbb{R}$.
So $(\mathbb{R}, \mathcal{T}_u)$ is metrizable space.
**Example 2:**
Consider $\mathbb{R}^2$ with usual topology. Recall usual metric

$$d(x,y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

on $\mathbb{R}^2$. Now the topology induced by $d$ on $\mathbb{R}^2$ is the usual topology on $\mathbb{R}^2$. So $(\mathbb{R}^2, \mathcal{T}_u)$ is metrizable space.

## Examples: Metrizable Spaces

**Example 3:**
Consider $X$ with discrete topology. Recall trivial metric.

$$d(x,y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}$$

where $x, y \in X$.
The topology induced by the above metric is the discrete topology on $X$. So discrete space is metrizable space.

## Topology

Examples:
Non-Metrizable Spaces

## Examples: Non-Metrizable Spaces

Let us see some examples of non-metrizable spaces, i.e. the topological space $(X, \mathcal{T})$ such that there exists no metric which induces $\mathcal{T}$.

## Examples: Non-Metrizable Spaces

**Example 1:**
Consider $X \neq \{x\}$ with indiscrete topology i.e. $\mathcal{T}_{ind} = \{\emptyset, X\}$. Then it is not metrizable space.

**Reason:**
It is because of the fact that in a metric space all the finite set are closed. But here only closed sects are $\emptyset$ and $X$. So $\mathcal{T}_{ind}$ can not be induced by any metric.

**Caution:**
It does not mean that if in some topological space all finite sets are closed then that space is metrizable.

## Examples: Non-Metrizable Spaces

**Example 2:**
Let $X$ be an infinite set. Consider cofinite topology on $X$. Then it is not metrizable space.

**Reason:**
There is a topological property called "Hausdorffness" which should exist in a metric space. So if there exists some metric corresponds to $(X, \mathcal{T}_{cof})$ then $(X, \mathcal{T}_{cof})$ should be Hausdorff but it is not true.
So $(X, \mathcal{T}_{cof})$ is not metrizable space.

**Caution:**
It does not mean that if some topological space is Hausdorff then that space is metrizable.

## Examples: Non-Metrizable Spaces

**Example 3:**
$\mathbb{R}$ with lower limit topology is not metrizable space.

**Reason:**
There are topological spaces called "Second Countable Spaces" and separable metric spaces are second countable. So if there exists some metric corresponds to $(\mathbb{R}, \mathcal{T}_l)$ then $(\mathbb{R}, \mathcal{T}_l)$ should be second countable but it is not true.
So $(\mathbb{R}, \mathcal{T}_l)$ is not metrizable space.

## Topology

First Countable Spaces

## First Countable Spaces

**Def:**
A topological space $(X, \mathcal{T})$ is called a "First Countable Space" if for each $p \in X$ there exists a countable class $\mathcal{B}_p$ of open subsets of $X$ containing $p$ such that for every $U_p \in \mathcal{T}$ (open set containing $p$) there exists $B_p \in \mathcal{B}_p$ such that $p \in B_p \subset U_p$.

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## First Countable Spaces

**Def:**
A topological space $(X, \mathcal{T})$ is called a "First Countable Space" if for each $p \in X$ there exists a countable local base at $p$.

## First Countable Spaces

**Remark:**
Every subspace of a first countable space is first countable.

$A \subset X$ ($X$ is first countable)
$\Rightarrow$ $A$ is also first countable.

Proof: Let $X$ be a first countable space and $A \subset X$, Let $p \in A$ $\therefore X$ is first countable.
So $\exists$ a countable local base at $p$.
$\mathcal{B}_p = \{B_n \mid n \in \mathbb{N}^*\}$.
Now $\mathcal{B}_{p_A} = \{A \cap B_n \mid n \in \mathbb{N}^*\}$

## First Countable Spaces

Now $\mathcal{B}_{p_A}$ is a local base at $p$. for $\mathcal{T}_A$
And $\mathcal{B}_{p_A}$ is countable So
$A$ is first countable. □

## Topology

**Examples:**
**First Countable Spaces**

## Examples: First Countable Spaces

Let us discuss some examples of first countable spaces.
**Example 1:**
Discrete space is first countable.
Remember a local base at a point $p$ in discrete space, i.e.

$$\mathcal{B}_p = \{\{p\}\}$$

and it is finite.

## Examples: First Countable Spaces

**Example 2:**
Let $X = \{a, b, c, d\}$ and

$$\mathcal{T} = \{\emptyset, \{a\}, \{b\}, \{a, b\}, \{c, d\}, \{a, c, d\}, \{b, c, d\}, X\}$$

be a topology defined on $X$.

- $\mathcal{B}_a = \{\{a\}\}$
- $\mathcal{B}_b = \{\{b\}\}$
- $\mathcal{B}_c = \{\{c, d\}\}$
- $\mathcal{B}_d = \{\{c, d\}\}$

Remark: Every finite set with any top. is first countable.

## Examples: First Countable Spaces

**Example 3:**
$\mathbb{R}$ with usual topology is first countable.
For any real number $x$ one may consider the following countable local base at $x$.

$$\mathcal{B}_x = \left\{\left(x - \frac{1}{n}, x + \frac{1}{n}\right) \mid n \in \mathbb{Z}^+\right\}$$

## Examples: First Countable Spaces

**Example 3:**
$\mathbb{R}$ with usual topology is first countable.
For any real number $x$ one may consider the following countable local base at $x$.

$$\mathcal{B}_x = \left\{\left(x - \frac{1}{n}, x + \frac{1}{n}\right) \mid n \in \mathbb{Z}^+\right\}$$

$\mathcal{B}_x$ is countable
$\Rightarrow$ $(\mathbb{R}, \mathcal{T}_u)$ is first countable

## Examples: First Countable Spaces

**Example 4:**
Every metric space is first countable.

Sol: Let $X$ be a metric space $(X, d, \mathcal{T})$.
Now consider the following collection of open balls for $p \in X$.

$$\mathcal{B}_p = \{ B(p, \tfrac{1}{n}) \mid n \in \mathbb{N} \}.$$

$\mathcal{B}_p$ is a local base at $p$. $\mathcal{B}_p$ is countable
$\Rightarrow$ $X$ is first countable.



## Topology

Second Countable Spaces



## Second Countable Spaces

**Def:**
A topological space $(X, \mathcal{T})$ is called a "Second Countable Space" if there exists a countable class $\mathcal{B}$ of open subsets of $X$ such that for every $U \in \mathcal{T}$ and for all $x \in U$ there exists $B \in \mathcal{B}$ such that $x \in B \subset U$.



## Second Countable Spaces

**Def:**
A topological space $(X, \mathcal{T})$ is called a "Second Countable Space" if there exists a countable class $\mathcal{B}$ of open subsets of $X$ such that for every $U \in \mathcal{T}$ and for all $x \in U$ there exists $B \in \mathcal{B}$ such that $x \in B \subset U$.

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## Second Countable Spaces

**Def:**
A topological space $(X, \mathcal{T})$ is called a "Second Countable Space" if there exists a countable basis for $\mathcal{T}$.



## Second Countable Spaces

**Remark:**
Every subspace of a second countable space is second countable.

Proof: Let $A \subset X$ ($X$ is second countable space)
So $\exists$ $\mathcal{B} = \{ B_n \mid n \in \mathbb{N} \}$ countable.
Now $\mathcal{B}_A = \{ A \cap B_n \mid n \in \mathbb{N} \}$ is a basis for $\mathcal{T}_A$. And $\mathcal{B}_A$ is countable
$\Rightarrow$ $A$ is second countable. □



## Topology

Examples: Second Countable Spaces



## Examples: Second Countable

Let us see discuss some examples of second countable spaces.

**Example 1:**
Let $X$ be a countable set with discrete topology. Then $(X, \mathcal{T}_{Dis})$ is second countable. Recall basis for discrete space, i.e.

$$\mathcal{B} = \{\{x\} | x \in X\}$$

## Examples: Second Countable

**Example 2:**
Let $X = \{a, b, c, d\}$ and

$$\mathcal{T} = \{\emptyset, \{a\}, \{b\}, \{a, b\}, \{c, d\}, \{a, c, d\}, \{b, c, d\}, X\}$$

be a topology defined on $X$. A basis for $\mathcal{T}$ is

$$\mathcal{B} = \{\{a\}, \{b\}, \{c, d\}\}$$

## Examples: Second Countable

**Example 2:**
Let $X = \{a, b, c, d\}$ and

$$\mathcal{T} = \{\emptyset, \{a\}, \{b\}, \{a, b\}, \{c, d\}, \{a, c, d\}, \{b, c, d\}, X\}$$

be a topology defined on $X$. A basis for $\mathcal{T}$ is

$$\mathcal{B} = \{\{a\}, \{b\}, \{c, d\}\}$$

Remark: Any finite set with any top is second countable.

## Examples: Second Countable

**Example 2:**
Let $X = \{a, b, c, d\}$ and

$$\mathcal{T} = \{\emptyset, \{a\}, \{b\}, \{a, b\}, \{c, d\}, \{a, c, d\}, \{b, c, d\}, X\}$$

be a topology defined on $X$. A basis for $\mathcal{T}$ is

$$\mathcal{B} = \{\{a\}, \{b\}, \{c, d\}\}$$ finite.

basis

So $(X, \mathcal{T})$ is second countable.

## Examples: Second Countable

**Example 3:**
$\mathbb{R}$ with usual topology is second countable.
Consider the following basis for usual topology on $\mathbb{R}$.

$$\mathcal{B} = \{(a, b) \,|\, a, b \in \mathbb{Q}\}$$

## Examples: Second Countable

**Example 3:**
$\mathbb{R}$ with usual topology is second countable.
Consider the following basis for usual topology on $\mathbb{R}$.

$$\mathcal{B} = \{(a, b) \,|\, a, b \in \mathbb{Q}\}$$

Countable basis

So $(\mathbb{R}, \mathcal{T}_u)$ is second countable.

Ex: 4. $(\mathbb{R}, \mathcal{T}_{Dis})$ is not second countable.

$\mathcal{B}_{\mathcal{T}_{Dis}} = \{\{x\} \mid x \in \mathbb{R}\}$:

## Examples: Second Countable

**Example 5:**
$\mathbb{R}^2$ with usual topology is second countable.
Consider a basis $\mathcal{B}$ for usual topology on $\mathbb{R}^2$ consists of collection of open disks with rational radii and centers with rational coordinates. Such collection is countable.

$r, x, y \in \mathbb{Q}$

## Topology

**Second Countability implies First Countability**



## Second Countability implies...

- First Countability Axiom
- Second Countability Axiom

**Theorem:**
Let a topological space $(X, \mathcal{T})$ is second countable space. Then it also first countable space.

## Second Countability implies...

**Proof:** Let $X$ be a second countable space.
i.e $\exists$ $\mathcal{B}$ (a basis) s.t $\mathcal{B}$ is countable.
Now let $p \in X$ and consider
$\mathcal{B}_p \subset \mathcal{B}$ $\quad \mathcal{B}_p = \{B_p \mid p \in B_p\} \subset \mathcal{B}$
Now $\mathcal{B}_p$ is countable $\because$ $\mathcal{B}$ is countable
Now $\mathcal{B}_p$ is local base at $p$:

## Second Countability implies...

$\mathcal{B}_p \subset \mathcal{B}$ (basis) $\quad \forall\ U \in \mathcal{T}$ s.t $p \in U$
$\exists\ B_p \in \mathcal{B}_p$ s.t
$p \in B_p \subset U$
$\Rightarrow$ $\mathcal{B}_p$ is a local base at $p$.
And this exists $\forall$ $p \in X$.
$\Rightarrow$ $X$ is first countable $\square$

## Topology



First Countability
$\not\Rightarrow$ Second Countability

## First Countability $\not\Rightarrow$...

It has already been proved that every second countable space is first countable.
But what about its reveres implication, i.e.
For a first countable space what one can say about its second countability?
Before answering this question, let us first discuss some examples.

## First Countability $\not\Rightarrow$...

**Example 1:**
Let $X$ be an infinite set with discrete topology. Then it is first countable space. Since for any point $p \in X$.

$$\mathcal{B}_p = \{\{p\}\}$$

is a local base at $p$. And it is finite so it is first countable. But it is not second countable.

$$\mathcal{B} = \{\{x\} | x \in X\}$$

is the smallest basis of discrete space. Since $X$ is infinite so $\mathcal{B}$ is infinite.

## First Countability $\not\Rightarrow$...

**Example 2:**
Consider $\mathbb{R}$ with usual topology. It is both first countable and second countable.

**Example 3:**
Consider $\mathbb{R}$ with lower limit topology. It is first countable but not second countable.

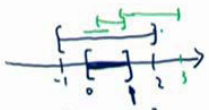
**Remark:**
A first countable space may or may not be second countable.

## Topology

Open Cover

## Open Cover

Let $X = [0,1]$




$\mathcal{U} = \{[-1, 2]\}$
$[0,1] \subset [-1, 2]$ $\mathcal{U}$ is cover of $[0,1]$
$\mathcal{V} = \{[0, \frac{1}{2}), (\frac{1}{4}, 1.9], (1.8, 3)\}$

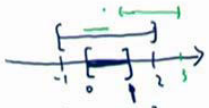
**Cover:**
A cover of a set $X$ is a collection of sets

$$\mathcal{U} = \{U_i | i \in I\}$$

where $I$ is index set, such that

$$X \subset \bigcup_{i \in I} U_i$$

## Open Cover



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$$\mathcal{U} = \{U_i | i \in I\}$$

where $I$ is index set, such that

$$X \subset \bigcup_{i \in I} U_i$$

## Open Cover

**Cover:**(In Topology)
Let $(X, \mathcal{T})$ be topological space. A cover of a $X$ is a collection of subsets of $X$, i.e.

$$\mathcal{U} = \{U_i | U_i \subset X, i \in I\}$$

where $I$ is index set, such that

$$X = \bigcup_{i \in I} U_i$$

## Open Cover

**Open Cover:**
Let $(X, \mathcal{T})$ be topological space. An open cover of a $X$ is a collection of open subsets of $X$, i.e.

$$\mathcal{U} = \{U_i | U_i \in \mathcal{T}, i \in I\}$$

where $I$ is index set, such that

$$X = \bigcup_{i \in I} U_i$$

## Open Cover

**Open Cover of Subset:**
Let $(X, \mathcal{T})$ be topological space and $Y \in X$. An open cover of a $Y$ is a collection of open subsets of $X$, i.e.

$$\mathcal{C} = \{C_i | C_i \in \mathcal{T}, i \in I\}$$

where $I$ is index set, such that

$$Y \subset \bigcup_{i \in I} C_i$$

## Topology

Examples: Open Cover

## Examples: Open Cover

Let us see some examples of open cover for the better understanding of the definitions we discussed in previous topic.

## Examples: Open Cover

**Example 1:**
Let $X = \{a, b, c, d\}$ with topology

$$\mathcal{T} = \{\phi, \{a\}, \{a, b\}, \{a, c, d\}, X\}$$

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Let $X = \{a, b, c, d\}$ with topology

$$\mathcal{T} = \{\phi, \{a\}, \{a, b\}, \{a, c, d\}, X\}$$

$\mathcal{U} = \{\{a, b\}, \{a, c, d\}\}$ open cover of $X$

## Examples: Open Cover

**Example 2:**
Let $X = \mathbb{R}$ with usual topology and consider the following collections of subset of $\mathbb{R}$

$$\mathcal{U} = \{(-n, n) | n \in \mathbb{N}\}$$

$$\mathcal{V} = \{(n, n+2) | n \in \mathbb{Z}\}$$

$$\mathcal{W} = \{(n, n+1) | n \in \mathbb{Z}\}$$

$$\mathcal{Y} = \{[n, n+1] | n \in \mathbb{Z}\}$$

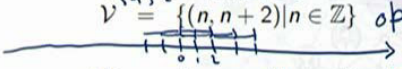


## Examples: Open Cover

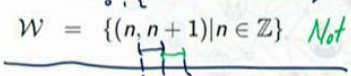
**Example 2:**
Let $X = \mathbb{R}$ with usual topology and consider the following collections of subset of $\mathbb{R}$

$\mathcal{U} = \{(-n, n) | n \in \mathbb{N}\}$ open cover

$\mathcal{V} = \{(n, n+2) | n \in \mathbb{Z}\}$ open cover



$\mathcal{W} = \{(n, n+1) | n \in \mathbb{Z}\}$ Not a cover of $\mathbb{R}$



$\mathcal{Y} = \{[n, n+1] | n \in \mathbb{Z}\}$ $\mathcal{Y}$ is a cover

## Examples: Open Cover

**Example 3:**
Let $X = \mathbb{R}$ with usual topology and consider $(0, 1) \subset \mathbb{R}$ and $[0, 1) \subset \mathbb{R}$. The following collections of subset of $\mathbb{R}$.

$$\mathcal{U} = \{(1/n, 1) | n \in \mathbb{N}^*\}$$

$$\mathcal{V} = \{(n, n+2) | n \in \mathbb{Z}\}$$

$\mathcal{V}$ is an open cover of $(0,1)$ but it is not an open cover of $[0,1)$

$\mathcal{U}$ is an open cover of $(0,1)$ and open cover of $[0,1)$.

## Topology

Open Subcover



## Open Subcover

**Subcover:**
Let $(X, \mathcal{T})$ be topological space. Consider $\mathcal{C}$ covers $X$, i.e.

$$\mathcal{C} = \{C_i | C_i \subset X\}$$

such that $X = \bigcup_i C_i$.
A subclass $\mathcal{D}$ of $\mathcal{C}$ that also covers $X$ is called an open subcover of $\mathcal{C}$.



## Open Subcover

**Open Subcover:**
Let $(X, \mathcal{T})$ be topological space. Consider $\mathcal{C}$ be an open cover of $X$, i.e.

$$\mathcal{C} = \{C_i | C_i \in \mathcal{T}\}$$

such that $X = \bigcup_i C_i$.
A subclass $\mathcal{D}$ of $\mathcal{C}$ that also covers $X$ is called an open subcover of $\mathcal{C}$.



## Open Subcover

**Example 1:**
Let $X = \{a, b, c, d\}$ with topology

$$\mathcal{T} = \{\phi, \{a\}, \{a, b\}, \{c, d\}, \{a, c, d\}, X\}$$

$\mathcal{C} = \{\{a\}, \{a, b\}, \{c, d\}\}$ is an open cover of $X$.

$\mathcal{D} = \{\{a, b\}, \{c, d\}\}$ it again covers $X$ so $\mathcal{D}$ is a subcover of $\mathcal{C}$

$\mathcal{U} = \{\{a\}, \{c, d\}\}$

## Examples: Open Cover

**Example 2:**
Let $X = \mathbb{R}$ with usual topology and consider

$$\mathcal{U} = \{(-n, n) | n \in \mathbb{N}\}$$

$\mathcal{D} \subset \mathcal{U}$

$\mathcal{D} = \{(-n, n) \mid n \in \mathbb{N} - \{1, 2, 3, 4, 5\}\}$


## Examples: Open Cover

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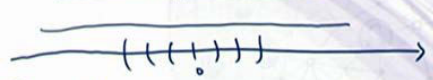


$\mathcal{D} \subset \mathcal{U}$

$\mathcal{D} = \{(-n, n) \mid n \in \mathbb{N} - \{1, 2, 3, 4, 5\}\}$


## Topology

Lindelöf Space


## Lindelöf Space

**Lindelöf Space:**
A topological space $(X, \mathcal{T})$ is called a Lindelöf space if and only if every open cover of $X$ has a countable subcover.

## Lindelöf Space

**Example 1:**
Let $X$ be a finite set with any topology then it is Lindelöf.

**Example 2:**
A set $X$ with indiscrete topology then it is Lindelöf.

**Example 3:**
$X = \mathbb{R}$ with usual topology is Lindelöf.


## Lindelöf Space


**Remark:**
A subspace of a Lindelöf space needs not be Lindelöf.

## Topology

Second Countable
and
Lindelof Spaces

## Second Countable...

**Theorem:**
Every second countable space is Lindelöf space.

## Second Countable...

Proof: Let $(X, \mathcal{T})$ be a second countable space. i.e $\exists$ a countable basis for $\mathcal{T}$

Say $\mathcal{B} = \{B_n\}$

Let $\mathcal{A}$ be an open cover of $X$.

$$X = \bigcup_i A_i \quad A_i \in \mathcal{A} \subset \mathcal{T}$$

Now for each positive integer "n" choose an element $A_n$ of $\mathcal{A}$ s.t $B_n \subset A_n$.

## Second Countable...

Now collect all such elements of $\mathcal{A}$

$$\mathcal{A}' = \{A_n \mid B_n \subset A_n\} \subset \mathcal{A}$$

Now $\mathcal{A}'$ is countable.

## Second Countable...

Now collect all such elements of $\mathcal{A}$

$$\mathcal{A}' = \{A_n \mid B_n \subset A_n\} \subset \mathcal{A}$$

Now $\mathcal{A}'$ is countable.

Now we have to show that $\mathcal{A}'$ covers $X$.

Let $x \in X$ and consider an element $A$ of $\mathcal{A}$ such $A$ exists $\because \mathcal{A}$ is a open cover.

Moreover this one $A$ is open

## Second Countable...

$x \in A$, $\exists$ $B_n \in \mathcal{B}$ (Basis)

s.t $x \in B_n \subset A$

But for $B_n$ $\exists$ $A_n \in \mathcal{A}'$

s.t $x \in B_n \subset A_n$

$\Rightarrow$ $X = \bigcup_i A_i$ $\quad A_i \in \mathcal{A}'$

$\Rightarrow$ $\mathcal{A}'$ covers $X$ So $X$ is Lindelöf.

## Topology

Separable Spaces



## Separable Spaces

**Def:**
A topological space $(X, \mathcal{T})$ is said to be "Separable" if there exists a countable dense subset $A$ of $X$.
i.e.
$\exists$ $A \subset X$ such that
1. $A$ is countable.
2. $\overline{A} = X$.



## Separable Spaces

**Def:**
A topological space $(X, \mathcal{T})$ is said to be "Separable" if there exists a sequence of points

$$\{x_n\}_{n=1}^{\infty}$$

in $X$ such that every nonempty subset of $X$ contains at least one point of the sequence.

## Separable Spaces

**Remark:**
Every second countable space is separable. The converse of this is not true in general.

**Counter Example:**
$\mathbb{R}$ with lower limit topology is separable but not second countable.

## Topology

Examples: Separable Spaces

## Examples: Separable Spaces

Let us discuss some examples of separable spaces.

**Example 1:**
Let $X$ be a finite set. Then $X$ with any topology is separable.

**Example 2:**
Consider a set $X$ with indiscrete topology. Then $X$ separable.

## Examples: Separable Spaces

**Example 3:**
$\mathbb{R}$ with usual topology is separable.

**Reason:**
Since there exists $\mathbb{Q}$ in $\mathbb{R}$ that is countable and

$$\overline{\mathbb{Q}} = \mathbb{R}.$$

**Example 4:**
$\mathbb{R}$ with discrete topology is not separable.

**Reason:**
Since there exists no countable subset in $\mathbb{R}$ that is dense in $\mathbb{R}$.

## Topology

Some More Examples: Separable Spaces

## Some More Examples

Let us discuss some more examples of separable spaces. We will discuss these examples in the form of propositions.

## Some More Examples

**Proposition 1:**
A discrete space is separable if and only if it is countable.

Proof: $X$, $\mathcal{T}_{dis} = P(X)$
We know that the only closed subset in $X$ is $X$ itself.
So the only choice for $A$ is $X$ s.t $\overline{A} = X$
$\Rightarrow$ $X$ is separable iff $X$ is countable. □

## Some More Examples

**Proposition 2:**
$(X, T_{cof})$ is separable.

Proof: If $X$ is finite $\Rightarrow \mathcal{T}_{cof} = \mathcal{T}_{dis}$
$\Rightarrow$ $X$ is separable.
Now Let $X$ is not finite.
Now consider $A \subset X$ (Countable (Not finite))

Recall $\mathcal{T}_{cof} = \{U \subset X \mid U^c = \text{finite on } X\}$

## Some More Examples

$B \subset X$ (closed) $\Rightarrow$ B is finite

Now recall the closure of a set is the smallest closed superset of the set. So $\overline{B} = B$
Now for $A$ the only closed superset of $A$ is $X$.
$\Rightarrow$ $\overline{A} = X$ $\Rightarrow$ $X$ is sep

## Topology

Subspace of Separable Space

## Subspace of Separable Space

As we proved that the subspace of a first countable space is first countable and the subspace of a second countable space is second countable. Now what about the subspace of a separable space? i.e.

Is subspace of a separable is separable?

Not in general.

## Subspace of Separable Space

**Example:**
Consider $\mathbb{R}^2$ with topology generated by

$$\mathcal{B} = \bigcup_{a,b,c,d \in \mathbb{R}} \{(x,y) | a \leq x < b, c \leq y < d\}.$$

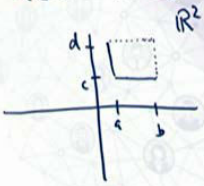
Now it is separable.

## Subspace of Separable Space

**Example:**
Consider $\mathbb{R}^2$ with topology generated by

$$\mathcal{B} = \bigcup_{a,b,c,d \in \mathbb{R}} \{(x,y) | a \leq x < b, c \leq y < d\}.$$

Now it is separable. $\{(x,y) \mid a \leq x < b,\ c \leq y < d\} = [a,b) \times [c,d)$ in $\mathbb{R}^2$




## Subspace of Separable Space

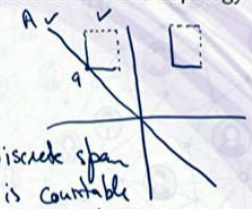
Consider

$$A = \{(x,y) | x = -y\} \subset \mathbb{R}^2$$

with relative topology on $A$ relative to the said topology on $\mathbb{R}^2$.

$\mathcal{T}_A = \{ -- , \{q\}, -- \}$

" Discrete top on A




And we know that Discrete space is separable iff A is countable
But here A is not countable.
$\therefore$ $(A, \mathcal{T}_A)$ is not separable.

## Topology

Metric Spaces and Separability

## Metric Spaces and Separability

We have seen that every metric space is first countable space but we did not discuss about the second countability and separability of metric spaces. In fact a metric space is not second countable and not separable in general.

**Example:**
An infinite set $X$ with trivial metric is not second countable and not separable.

## Metric Spaces and Separability

**Theorem:**
A separable metric space is second countable.

**Proof:** Now $X$ is separable. So $\exists\ A \subset_{\text{countable}} X$ s.t $\overline{A} = X$

Now consider a class of all open balls of the kind $B(a,r)$ where $a \in A$, $r \in \mathbb{Q}$

i.e $\mathcal{B} = \{B(a,r) \mid a \in A,\ r \in \mathbb{Q}\}$

$\because$ $A$ and $\mathbb{Q}$ both countable So $\mathcal{B}$ is:

## Metric Spaces and Separability

Now for showing $X$ is second countable we have to prove that $\mathcal{B}$ is a basis for the metric top. on $X$.

Let $U \in \overline{S}$ and $x \in U$

$\because$ $X$ is a metric space so

$\exists\ \varepsilon,\ B(x,\varepsilon)$ and $x \in B(x,\varepsilon) \subset U$ — (I)

$\because$ $A$ is dense in $X$ so $\exists$ a point $a \in A$ s.t $d(a,x) < \varepsilon$ Let $d(a,x) < \frac{1}{3}\varepsilon$.



## Metric Spaces and Separability

Now consider a rational no $\delta$ s.t.

$$\frac{1}{3}\varepsilon < \delta < \frac{2}{3}\varepsilon$$

and consider $B(a,\delta)$

$x \in B(a,\delta)$ — (II) $\because d(a,x) < \frac{1}{3}\varepsilon < \delta$

Now Let $b \in B(a,\delta) \Rightarrow d(a,b) < \delta$

$$d(x,b) \le d(x,a) + d(a,b)$$
$$< \frac{1}{3}\varepsilon + \delta < \frac{1}{3}\varepsilon + \frac{2}{3}\varepsilon = \varepsilon$$



## Metric Spaces and Separability

$\Rightarrow$ $B(a,\delta) \subset B(x,\varepsilon)$

By (I)

$x \in B(a,\delta) \subset B(x,\varepsilon) \subset U$

Countable $\mathcal{B}$ $a \in A$, $\delta \in \mathbb{Q}$

$\Rightarrow$ $\mathcal{B}$ is a basis for the metric sp $X$

$\Rightarrow$ $X$ is second countable.



## Topology

Separation Axioms



## Separation Axioms

1. $T_0$ Spaces
2. $T_1$ Spaces
3. $T_2$ (Hausdorff) Spaces
4. Regular Spaces
5. $T_3$ Spaces
6. Normal Spaces
7. $T_4$ Spaces



## Topology

$T_0$ Space



## $T_0$ Space

**Def:**
A topological space $(X, T)$ is said to be "$T_0$ Space" iff for each $x, y \in X$ such that $x \neq y$ there exists an open set $U_x \subset X$ containing $x$ such that $y \notin U_x$

or

there exists an open set $V_y \subset X$ containing $y$ such that $x \notin V_y$.

Dr. Hani Shakir



## $T_0$ Space

**Remark:**
$T_0$ property is a topological property.

Proof: Let $X$ be space that is $T_0$

Consider $f: X \longrightarrow Y$ $f$ is bijection + Continuous $f^{-1}$ is cont.

Now $x, y \in X$ $x \neq y$

$\exists\ U_x \ni x$ (open set)

s.t $y \notin U_x$ Now $f(U_x) \subset$ open (Y)

moreover $f(y) \notin f(U_x)$

$\Rightarrow$ $Y$ is $T_0$.

## Topology

Examples: $T_0$ Space

## Examples: $T_0$ Space

Let us discuss some examples of $T_0$ spaces.

**Example 1:**
Every discrete space is $T_0$.

Example: 2:
Let $X \neq \{a\}$
Consider $\mathcal{T}_{indis} = \{\phi, X\}$
$(X, \mathcal{T}_{indis})$ is not $T_0$

## Examples: $T_0$ Space

**Example 3:** Consider $X = \{a, b, c, d\}$ with

$$\mathcal{T} = \{\phi, \{a\}, \{a, b\}, \{a, c\}, \{a, b, c\}, X\}$$

## Examples: $T_0$ Space

**Example 3:** Consider $X = \{a, b, c, d\}$ with

$$\mathcal{T} = \{\phi, \{a\}, \{a, b\}, \{a, c\}, \{a, b, c\}, X\}$$

$(X, \mathcal{T})$ is $T_0$.

## Examples: $T_0$ Space

**Example 4:** Consider $X = \{a, b, c, d\}$ with

$$\mathcal{T} = \{\phi, \{a, c\}, \{b, d\}, X\}$$

## Examples: $T_0$ Space

**Example 4:** Consider $X = \{a, b, c, d\}$ with

$$\mathcal{T} = \{\phi, \{a, c\}, \{b, d\}, X\}$$

So $(X, \mathcal{T})$ is not $T_0$

## Topology

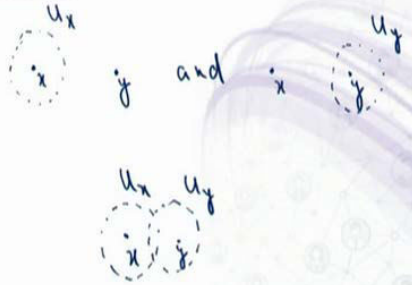
$T_1$ Space

## $T_1$ Space

**Def:**
A topological space $(X, \mathcal{T})$ is said to be "$T_1$ Space" iff for each $x, y \in X$ such that $x \neq y$ there exist open subsets $U_x, U_y$ of $X$ containing $x, y$ respectively such that $y \notin U_x$ and $x \notin U_y$.

Dr. Hani Shakir

## $T_1$ Space






## $T_1$ Space

**Remark:**
$T_1$ property is a topological property.



## Topology

Examples: $T_1$ Space

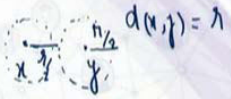


## Examples: $T_1$ Space

Let us discuss some examples of $T_1$ spaces.

**Example 1:**
Every discrete space is $T_1$.

**Example 2:**
Every metric space is $T_1$.






## Examples: $T_1$ Space

**Example 3:** Every cofinite space is $T_1$.

Sol Let $(X, \mathcal{T}_{cof})$
Let $X$ is finite.
$\mathcal{T}_{cof} = P(X)$ discrete.
If $X$ is not finite. Let $U \in \mathcal{T}_{cof}$
iff $U = \phi$ or $U^c$ = finite.
$x, y \in X$ $x \in U_x = X \setminus \{y\}$ $y \in U_y = X \setminus \{x\}$
but $y \notin U_x$ but $x \notin U_y$



## Examples: $T_1$ Space

**Example 4:** Consider $X = \{a, b, c, d\}$ with

$$\mathcal{T} = \{\phi, \{a\}, \{b\}, \{a, b\}, X\}$$



## Examples: $T_1$ Space

**Example 4:** Consider $X = \{a, b, c, d\}$ with

$$\mathcal{T} = \{\phi, \{a\}, \{b\}, \{a, b\}, X\}$$

$a, b \in \{c\}^c$, $\{b\}^c$ but $a \notin \{b\}^c$

$c, d$

$(X, \mathcal{T})$ is not $T_1$.



## Topology

$T_0 \not\Longrightarrow T_1$

## $T_0 \not\Longrightarrow T_1$

$T_1$ property is much stronger property than $T_0$ property.

i.e.

Every $T_1$ space is $T_0$ but not every $T_0$ space is $T_1$.
Let us prove first part of the claim and give an example for the second part.



## $T_0 \not\Longrightarrow T_1$

$T_1 \Rightarrow T_0$
$\not\Leftarrow$



## $T_0 \not\Longrightarrow T_1$

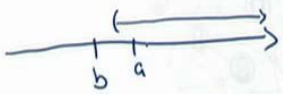
**Example:** A $T_0$ space that is not $T_1$.
Consider $\mathbb{R}$ with $\mathcal{T}$ generated by

$$\{(a,\infty) | a \in \mathbb{R}\}$$

$a, b \in \mathbb{R}$ $\quad a > b$

$b < c < a$

$a \in U_a = (c, \infty) \not\ni b$ So $T_0$-Space






## Topology

Criteria for $T_1$ Space



## Criteria for $T_1$ Space

We have discussed $T_1$ space and its examples.
Now we will state and prove a very simple criteria to check $T_1$ property in some space.



## Criteria for $T_1$ Space

**Theorem:**
A topological space $(X, \mathcal{T})$ is $T_1$ iff every singleton subset $\{x\} \subset X$ is closed.

Proof: Let $(X, \mathcal{T})$ is $T_1$ space
Now we have to show that $\forall\, x \in X$
$\{x\} \subset X$ is closed $\Rightarrow \{x\}^c \in \mathcal{T}$
Let $y \in \{x\}^c$ so $x \neq y$
So $\exists\, U_y$ open s.t $x \notin U_y$ $\because X$ is $T_1$



## Criteria for $T_1$ Space

$\Rightarrow \quad y \in U_y \subset \{x\}^c$
$\Rightarrow \{x\}^c$ is open in $X$
$\Rightarrow \{x\}$ is closed in $X$

For the other side.
Let us each singleton subset $\{x\} \subset X$ is closed.



## Criteria for $T_1$ Space

$x \neq y \in X$
$x \in U_x = X \setminus \{y\} = \{y\}^c \in \mathcal{T}$
$y \in U_y = X \setminus \{x\} = \{x\}^c \in \mathcal{T}$

$\Rightarrow (X, \mathcal{T})$ is a $T_1$ space

## Topology

Subspace of $T_1$ Space

## Subspace of $T_1$ Space

In this topic we will discuss about the subspace of $T_1$ space. In fact the main aim here is to show that every subspace of a $T_1$ space is $T_1$.

## Subspace of $T_1$ Space

**Theorem:**
Every subspace of a $T_1$ space is also a $T_1$ space.

Proof: Let $(X,\mathcal{T})$ be a $T_1$ space $Y \subset X$ and $(Y,\mathcal{T}_Y)$ be a subspace of $(X,\mathcal{T})$
We have already discussed that if in a space every singleton subset is closed then that space is $T_1$
Let $p \in Y \subset X$ $\because$ X is $T_1$.
So $\{p\} \subset X$ closed.

## Subspace of $T_1$ Space

Ta $\{p\}^c$ is open in X
So $\rightarrow$ $Y \cap \{p\}^c$ is open in Y
But $Y \cap \{p\}^c = Y \setminus \{p\}$
$\Rightarrow$ $\{p\}$ is closed in Y
$\Rightarrow (Y, \mathcal{T}_Y)$ is $T_1$.

## Topology

Properties of $T_1$ Spaces: I

## Properties of $T_1$ Spaces: I

Let us discuss some properties of $T_1$ spaces. Some of these properties can also be used as a criterion of $T_1$ property.

**Property I:** Every finite subset of $T_1$ space is closed.

## Properties of $T_1$ Spaces: I

Proof: Let $(X,\mathcal{T})$ be a $T_1$ space
$A \subset X$ finite Let $A = \{a_1, \cdots, a_n\}$
$A = \bigcup_{i=1}^{n} \{a_i\}$ $a_i \in A$ $\forall i$
$\because$ X is $T_1$ So every singleton subset of X is closed in X.
$\{a_i\}$ is closed $\forall$ $i = 1, 2, \cdots, n$.
$\Rightarrow$ A is a finite union of closed subset
$\Rightarrow$ A is closed in X.

## Properties of $T_1$ Spaces: I

**Property II:**
Let $X$ be a finite set. Then $(X,\mathcal{T})$ is a $T_1$ space if and only if $(X,\mathcal{T})$ is discrete space.

Proof: Let X is finite and $(X,\mathcal{T})$ is $T_1$ space. So in X every finite subset is closed.
$\Rightarrow$ Every subset of X is also open in X
$\Rightarrow \mathcal{T} = P(X)$ $\Rightarrow$ X is discrete.
Now If $(X,\mathcal{T})$ is discrete $\Rightarrow (X,\mathcal{T})$ is $T_1$

## Topology

Properties of $T_1$ Spaces: II

## Properties of $T_1$ Spaces: II

**Property III:**
Let $A$ be a finite subset of a $T_1$ space $X$ then $A'$ is empty.
**Proof:**
Let $A$ is finite subset of $X$, say $A$ has $n$ elements, i.e.

$$\{a_1, \ldots, a_n\}$$

Since $A$ is finite and $X$ is a $T_1$ space and we know that every finite subset of a $T_1$ space is closed so

$$A' \subset A$$

Now let us check one by one the points of $A$ as limits points of $A$.

## Properties of $T_1$ Spaces: II

Let's start with $a_1$.
Now note that

$$U_{a_1} = \{a_2, \ldots, a_n\}^c$$

is an open set containing $a_1$ and

$$A \cap (U_{a_1} \setminus \{a_1\}) = \emptyset$$

So $a_1$ is not a limit point of $A$.
Similarly, no other point of $A$ is a limit point of $A$. So

$$A' = \emptyset$$

## Topology

$T_2$/Hausdorff Spaces

## $T_2$/Hausdorff Spaces

**Def:**
A topological space $(X, \mathcal{T})$ is said to be "$T_2$ Space" iff for every distinct pair of elements $x, y \in X$ there exist open subsets $U_x, U_y$ of $X$ containing $x, y$ respectively such that $U_x \cap U_y = \emptyset$.

$T_2$ spaces are also called Hausdorff Spaces.

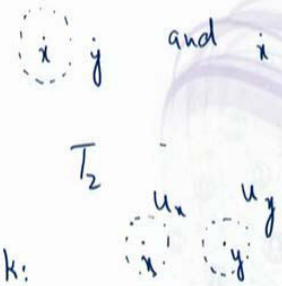
## $T_2$/Hausdorff Spaces

Felix Hausdorff
(1868 - 1942)

**Def:**
A topological space $(X, \mathcal{T})$ is said to be "$T_2$ Space" iff for every distinct pair of elements $x, y \in X$ there exist open subsets $U_x, U_y$ of $X$ containing $x, y$ respectively such that $U_x \cap U_y = \emptyset$.

$T_2$ spaces are also called Hausdorff Spaces.

## $T_2$/Hausdorff Spaces




Remark:
$T_2$ property is topological Prop.

## Topology

Examples: Hausdorff Spaces

## Examples: Hausdorff Spaces

In this topic we will discuss some examples of Hausdorff spaces.

**Example 1:** Every discrete space is Hausdorff.

**Example 2:** A nonempty set $X \neq \{x\}$ with indiscrete topology is not Hausdorff.

## Examples: Hausdorff Spaces

**Example 3:** $\mathbb{R}$ with lower limit topology is Hausdorff.

$[a,b) \quad (a,b)$

$x, y \in \mathbb{R} \quad x < y$

$x \in [x,c), [y,d) \ni y \quad x < c < y < d$

$U_x \cap U_y = \emptyset$

## Examples: Hausdorff Spaces

**Example 4:** $\mathbb{R}$ with topology generated

$$\{(a,\infty) | a \in \mathbb{R}\}$$

is not Hausdorff.

$3, 6 \in \mathbb{R}$

$6 \in (5,\infty)$

$3 \in (a,\infty) \not\ni 6 \quad a < 3$

## Examples: Hausdorff Spaces

**Example 5:** An infinite set with cofinite topology is not Hausdorff space.

## Topology

**Subspace of Hausdorff Space**

## Subspace of Hausdorff Space

In this topic we will discuss about Hausdorffness of a subspace of a Hausdorff space. In fact the main aim here is to show that the property of being a Hausdorff space is inherently.

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## Subspace of Hausdorff Space

**Theorem:**
Let $(X, \mathcal{T})$ is a Hausdorff space and $A \subset X$ then $(A, \mathcal{T}_A)$ is Hausdorff.

Proof: Let $A \subset X$

Given $(X, \mathcal{T})$ is Hausdorff.

$\Rightarrow \forall\ x, y \in X \quad x \neq y$

$\exists\ U_x, U_y \in \mathcal{T} \Rightarrow U_x \cap U_y = \emptyset$

$x \in U_x, \ y \in U_y$

$a, b \in A \subset X \quad \exists\ a \in U_a, b \in U_b \in \mathcal{T}$ s.t

$U_a \cap U_b = \emptyset$

## Subspace of Hausdorff Space

Now $U_a' = A \cap U_a$

$U_b' = A \cap U_b \in \mathcal{T}_A$

Now $a \in U_a', \quad b \in U_b'$

$U_a' \cap U_b' = (A \cap U_a) \cap (A \cap U_b)$

$= A \cap (U_a \cap U_b)$

$= A \cap (\emptyset) = \emptyset$

$\Rightarrow (A, \mathcal{T}_A)$ is Hausdorff. $\square$

## Topology

Metric Spaces are Hausdorff

## Metric Spaces are Hausdorff

Recall the topic where we discussed some examples of "Non Metrizable Spaces". We discussed there that an infinite set $X$ with cofinite topology is not metrizable. It is because of non-Hausdorffness of $X$ with respect to cofinite topology. In this topic we will discuss an important property of metric spaces called "Hausdorffness".

## Metric Spaces are Hausdorff

**Theorem:**
Every metric space is Hausdorff.

Proof $(X,d) \longrightarrow (X,\mathcal{T}_d)$

Let $x, y \in X \quad x \neq y$

$\because$ $X$ is metric space. $d(x,y) \neq 0$

$d(x,y) = r$ say.

Now $U_x = B(x, r/2)$, $U_y = B(y, r/2)$

My claim is $U_x \cap U_y = \emptyset$

## Metric Spaces are Hausdorff

Ld it is not true. So

$\exists \; z \in U_x \cap U_y$

Now $d(x,y) \leq d(x,z) + d(z,y)$

$< r/2 + r/2 = r = d(x,y)$

$\lightning$

So $\nexists \; z \in U_x \cap U_y$

$\Rightarrow U_x \cap U_y = \emptyset \Rightarrow (X,\mathcal{T}_d)$ is Hausdorff. $\square$

## Topology

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$T_1 \not\Longrightarrow T_2$

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## $T_1 \not\Longrightarrow T_2$

$T_2$ property is a stronger property than $T_1$ property.

i.e.

Every $T_2$ space is $T_1$ but not every $T_1$ space is $T_2$.

## $T_1 \not\Longrightarrow T_2$

**Example:** An infinite set $X$ with cofinite topology is $T_1$ space but not $T_2$ space.
**Justification:** Given $X$ is infinite. Now with respect to cofinite topology on $X$, a subset $D$ of $X$ is open in $X$ iff

$$D = \emptyset \quad \text{or} \quad D^c = \text{finite}$$

So for each pair of distinct points $x, y$ in $X$ one may consider

$$U_x = \{y\}^c \quad \text{and} \quad U_y = \{x\}^c$$

and it is clear that

$$x \in U_x \not\ni y \quad \text{and} \quad y \in U_y \not\ni x$$

## $T_1 \not\Longrightarrow T_2$

So $(X, \mathcal{T}_{cof})$ is $T_1$.
Now we will show that $(X, \mathcal{T}_{cof})$ is not $T_2$.
Lets start contrarily i.e. with the assumption $(X, \mathcal{T}_{cof})$ is $T_2$. So for any distinct pair of points $x, y$ in $X$, there exist $U_x, U_y$ (open subsets of $X$ containing $x$ and $y$ resp.), s.t.

$$U_x \cap U_y = \emptyset$$

Now $(U_x)^c$ and $(U_y)^c$ are finite.Now

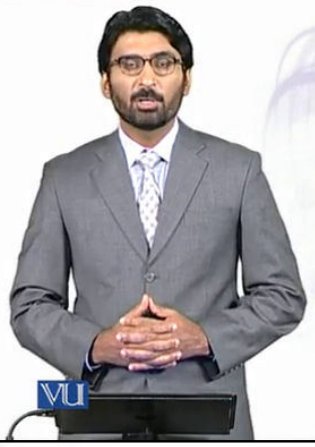
$$(U_x \cap U_y)^c = (U_x)^c \cup (U_y)^c$$

is finite. But $(U_x \cap U_y)^c = X$ that is infinite.
Contradiction. So $(X, \mathcal{T}_{cof})$ is not $T_2$.

## Topology

Unique Limit Point

## Unique Limit Point



Recall the definition of "Convergent Sequence". Let $(X, \mathcal{T})$ be a topological space. A sequence of points $(x_i)_{i \in I}$ in $X$ is said to be converges to a point $x \in X$ if for every open subset $U_x$, containing $x$, in $X$ there exists $N \in \mathbb{N}$ such that for all $n \geq N$, $x_n \in U_x$.

## Unique Limit Point

Now if we consider convergent sequences in $\mathbb{R}^n$ with usual topology then our experience for such sequences is uniqueness of the limit points.
But if we critically analyze the definition of convergent sequences, we find no such condition on the uniqueness of the limit point.

## Unique Limit Point

**Example:**
Consider $X = \{a, b, c\}$ with

$$\mathcal{T} = \{\emptyset, \{b\}, \{a, b\}, \{b, c\}, X\}$$

The constant sequence $(b)$ converges to $a$, to $b$, and to $c$.

## Unique Limit Point

Topological spaces in which sequences converge to unique limit points are of more importance than the topological spaces in which limit are not unique. And the reason is, very limited results can be stated/proved in spaces with multiple limit points. So one should impose some extra condition/property on the space to avoid such multiplicity. And this condition/property brings the class of spaces under consideration closer to which one's geometric intuition applies.
The property that ensure the uniqueness of limit point is the Hausdorff property.

## Topology

**Theorem: Unique Limit Point**

## Theorem: Unique Limit Point

**Theorem:**
Let $X$ be a Hausdorff space then every convergent sequence of points has a unique limit point.
i.e.
for $(x_i)$ in $X$ if

$$a, b \in X$$

are the limit points of $(x_i)$ then

$$a = b.$$

## Theorem: Unique Limit Point

**Proof:**
Consider a convergent sequence $(x_i)$ in a Hausdorff space $X$. Let $(x_i)$ converges to $a, b \in X$ and

$$a \neq b$$

Since $X$ is Hausdorff. So for $a, b \in X$ $\exists$ a pair of open subsets, $U_a$ and $V_b$, of $X$, such that

$$U_a \cap V_b = \emptyset$$

Now if $a$ is the limit of $(x_i)$ then

$\exists\ N \in \mathbb{N}$ such that $\forall\ n \geq N \Rightarrow x_n \in U_a$

## Theorem: Unique Limit Point

And if $b$ is the limit of $(x_i)$ then

$$\exists\ M \in \mathbb{N} \text{ such that } \forall\ n \geq M \ \Rightarrow\ x_n \in V_b$$

So it means that $\forall\ n \geq \max\{M, N\}$ we have

$$x_n \in U_a \cap V_b$$

Which is the contradiction of the fact

$$U_a \cap V_b = \emptyset$$

So $a = b$, i.e. in a Hausdorff space then every convergent sequence of points has a unique limit point.



## Topology

Regular Spaces

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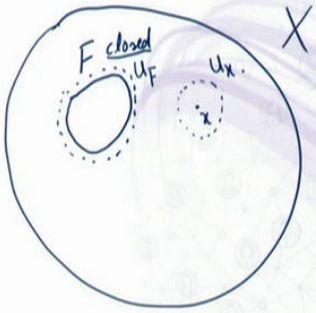
## Regular Spaces

**Def:**
A topological space $(X, \mathcal{T})$ is said to be "regular" iff for every closed subset $F$ of $X$ and for every $x \in F^c$ there exist open subsets $U_F$ and $U_x$ containing $F$ and $x$ respectively such that

$$U_F \cap U_x = \emptyset.$$



## Regular Spaces






## Regular Spaces

**$T_3$ Space:**
A topological space $(X, \mathcal{T})$ is said to be "$T_3$ space" iff it is either
Regular+$T_0$
or
Regular+$T_1$
or
Regular+$T_2$



## Regular Spaces

**Remarks:**

1. Property of being regular is a topological property.
2. Every subspace of a regular space is a regular space.
3. $T_3$ property is a topological property.
4. Every subspace of a $T_3$ space is a $T_3$ space.



## Topology

Examples: Regular Spaces



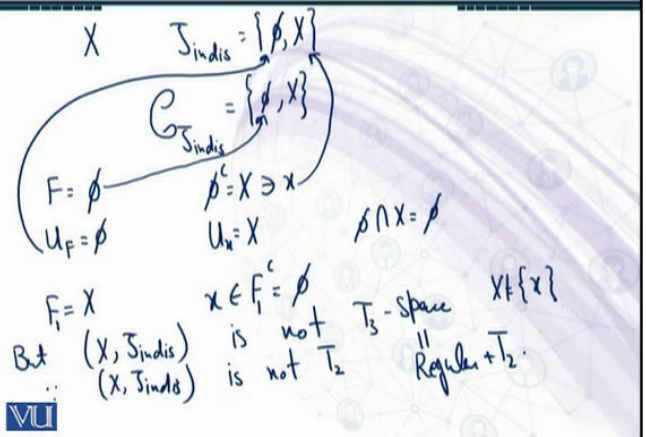
## Examples: Regular Spaces

In this topic we will discuss some examples of regular spaces.

**Example 1:**
Discrete space is a regular space.

**Example 2:**
Indiscrete space is a regular space.

## Examples: Regular Spaces



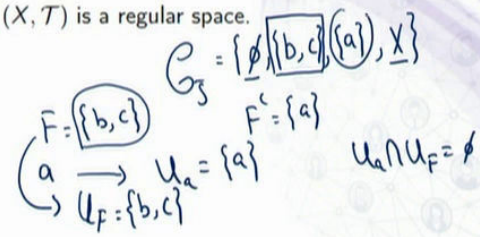



## Examples: Regular Spaces

**Example 3:** Consider $X = \{a, b, c\}$ with

$$\mathcal{T} = \{\emptyset, \{a\}, \{b, c\}, X\}.$$

$(X, \mathcal{T})$ is a regular space.



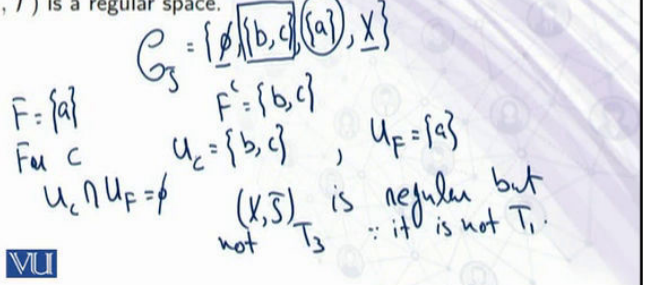



## Examples: Regular Spaces

**Example 3:** Consider $X = \{a, b, c\}$ with

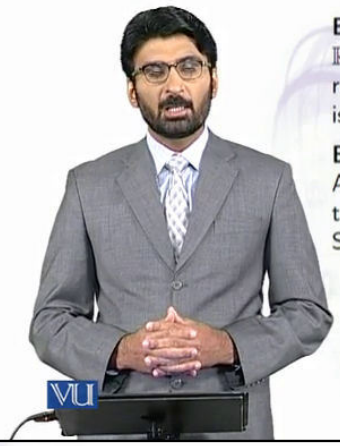
$$\mathcal{T} = \{\emptyset, \{a\}, \{b, c\}, X\}.$$

$(X, \mathcal{T})$ is a regular space.






## Examples: Regular Spaces



**Example 4:**
$\mathbb{R}$ with usual topology is regular. Moreover this space is also $T_3$ space.

**Example 5:**
An infinite set with cofinite topology is not regular. So it is not $T_3$.



## Examples: Regular Spaces

**Example 6:**
Recall the $K$-Topology on real line. Topology $\mathcal{T}_K$ generated by

$$\mathcal{B}_K = \{(a, b) |\ a, b \in \mathbb{R}\} \cup \{(a, b) \backslash K | a, b \in \mathbb{R}\}$$

where

$$K = \{1, 1/2, 1/3, \cdots\} \subset \mathbb{R}$$

$(\mathbb{R}, \mathcal{T}_K)$ is not a regular space.



## Topology

Normal Spaces



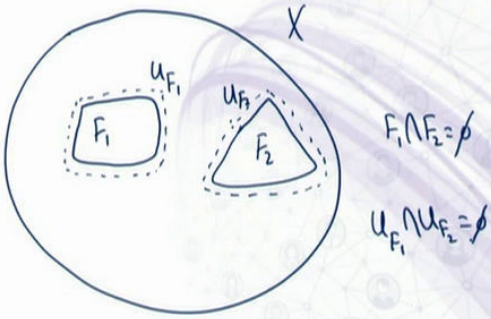
## Normal Spaces

**Def:**
A topological space $(X, \mathcal{T})$ is said to be "normal" iff for every pair of disjoint closed subsets, $F_1, F_2 \subset X$ there exist open subsets $U_{F_1}$ and $U_{F_2}$ containing $F_1$ and $F_2$ respectively such that

$$U_{F_1} \cap U_{F_2} = \emptyset.$$



## Normal Spaces



## Normal Spaces

**$T_4$ Space:**

A topological space $(X, \mathcal{T})$ is said to be "$T_4$ space" iff it is either
Normal+$T_1$
or
Normal+$T_2$

## Normal Spaces

**Remarks:**

1. Property of being normal is a topological property.
2. Subspace of a normal space needs not be a normal space.
3. $T_4$ property is a topological property.

**Completely Normal Space:**
A topological space $(X, \mathcal{T})$ is said to be completely normal space iff every subspace of $X$ with subspace topology is normal.

## Topology

Examples: Normal Spaces

## Examples: Normal Spaces

Let us see some examples for the better understanding of the definitions of the normal-axiom and related axioms.

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## Examples: Normal Spaces

**Example 1:** Consider $X = \{a, b, c, d\}$ with

$$\mathcal{T} = \{\emptyset, \{d\}, \{b, d\}, \{c, d\}, \{b, c, d\}, X\}.$$

**Claim:** $(X, \mathcal{T})$ is a normal space.
**Explanation:**
The closed collection is

$$\mathcal{C}_{\mathcal{T}} = \{\emptyset, \{a\}, \{a, b\}, \{a, c\}, \{a, b, c\}, X\}$$

## Examples: Normal Spaces

**Example 1:** Consider $X = \{a, b, c, d\}$ with

$$\mathcal{T} = \{\emptyset, \{d\}, \{b, d\}, \{c, d\}, \{b, c, d\}, X\}.$$

**Claim:** $(X, \mathcal{T})$ is a normal space. But $(X, \mathcal{T})$ is not $T_1$
**Explanation:**
The closed collection is

$$\mathcal{C}_{\mathcal{T}} = \{\emptyset, \{a\}, \{a, b\}, \{a, c\}, \{a, b, c\}, X\}$$

$F_1 = \emptyset$ $F_2 = X$ $U_{F_1} = \emptyset$, $U_{F_2} = X$
$U_{F_1} \cap U_{F_2} = \emptyset$

## Examples: Normal Spaces

**Example 2:** Consider $X = \{a, b, c, d\}$ with

$$\mathcal{T} = \{\emptyset, \{d\}, \{b, d\}, \{c, d\}, \{b, c, d\}, X\}.$$

**Claim:** $(X, \mathcal{T})$ is not $T_4$ space.

## Examples: Normal Spaces

**Example 3:** Consider $Y = \{b, c, d\}$ with

$$\mathcal{T}_Y = \{\emptyset, \{d\}, \{b, d\}, \{c, d\}, Y\}.$$

**Claim:** $(Y, \mathcal{T}_Y)$ is not a normal space.
**Explanation:**
The closed collection is

$$\mathcal{C}_{\mathcal{T}_Y} = \{\emptyset, \{b\}, \{c\}, \{b, c\}, Y\}$$

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## Examples: Normal Spaces

**Example 3:** Consider $Y = \{b, c, d\}$ with

$$\mathcal{T}_Y = \{\emptyset, \{d\}, \{b, d\}, \{c, d\}, Y\}.$$

**Claim:** $(Y, \mathcal{T}_Y)$ is not a normal space.
**Explanation:**
The closed collection is

$$\mathcal{C}_{\mathcal{T}_Y} = \{\emptyset, \{b\}, \{c\}, \{b, c\}, Y\}$$

$F_1 = \{b\}$, $F_2 = \{c\}$ $\quad U_{F_1} \cap U_{F_2} \neq \emptyset$



## Examples: Normal Spaces

**Example 4:** Consider $X = \{a, b, c, d\}$ with

$$\mathcal{T} = \{\emptyset, \{d\}, \{b, d\}, \{c, d\}, \{b, c, d\}, X\}.$$

**Claim:** $(Y, \mathcal{T}_Y)$ is not a completely normal space.
**Explanation:**
Consider $Y = \{b, c, d\}$. So the subspace topology on $Y$ with respect to $X$ is

$$\mathcal{T}_Y = \{\emptyset, \{d\}, \{b, d\}, \{c, d\}, Y\}.$$



## Examples: Normal Spaces

**Example 4:** Consider $X = \{a, b, c, d\}$ with

$$\mathcal{T} = \{\emptyset, \{d\}, \{b, d\}, \{c, d\}, \{b, c, d\}, X\}.$$

**Claim:** $(X, \mathcal{T}_Y)$ is not a completely normal space.
**Explanation:**
Consider $Y = \{b, c, d\}$. So the subspace topology on $Y$ with respect to $X$ is

$$\mathcal{T}_Y = \{\emptyset, \{d\}, \{b, d\}, \{c, d\}, Y\}.$$



## Examples: Normal Spaces

**Example 5:** $\mathbb{R}$ with usual topology is a $T_4$ space.

$(\mathbb{R}, \mathcal{T}_u)$ is Normal Space.

$F_1$ $\quad F_2$

$[a, b] \cap [c, d] = \emptyset$

$U_{F_1} = (a', b')$

$U_{F_2} = (b', d')$

$a' < a$

$b < b' < d$

$d' > d$

$U_{F_1} \cap U_{F_2} = \emptyset$



## Topology

Metric Spaces are Normal

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## Metric Spaces are Normal

In this topic we will discuss another important property of metric spaces, that is called "Normal Property". In fact more than this one metric spaces are $T_4$ space.

**Theorem:**
Every metric space is normal.



## Metric Spaces are Normal

**Definition:**
Let $(X, d)$ be a metric space and $A$ be a nonempty subset of $X$. Then a continuous function

$$d_A : X \longrightarrow \mathbb{R}$$

defined as

$$d_A(x) = \inf\{d(x, a) | a \in A\},$$

provides the distance between $x$ and $A$. We denote it as

$$\text{dist}(x, A) = d_A(x)$$

## Metric Spaces are Normal

**Proof:**
Let $(X, d)$ be a metric space and $F_1$ and $F_2$ be two disjoint closed subsets of $X$. Consider a subset $V_{F_1}$ of $X$ consists of all those elements $x$ of $X$ whose distance from $F_1$ is less than the distance from $F_2$, i.e.

$$V_{F_1} = \{x \in X | d_{F_1}(x) < d_{F_2}(x)\}$$

It is easy to see that $F_1 \subset V_{F_1}$.
Since $x \in F_1$ implies that $d_{F_1}(x) = 0$ and $d_{F_2}(x) > 0$, $F_2$ is closed.
So $d_{F_1}(x) < d_{F_2}(x)$ and we have $x \in V_{F_1}$.

## Metric Spaces are Normal

Now define a map

$$\delta = d_{F_2} - d_{F_1} : X \longrightarrow \mathbb{R}$$

Under this map

$$V_{F_1} = \delta^{-1}((0, \infty))$$

So $V_{F_1}$ is an open subset of $X$ containing $F_1$, since $\delta$ is continuous. On the same lines we can show that

$$V_{F_2} = \{x \in X | d_{F_2}(x) < d_{F_1}(x)\}$$

is an open subset of $X$ containing $F_2$. Moreover

$V_{F_1} \cap V_{F_2} = \emptyset \quad \Longrightarrow \quad X$ is normal.

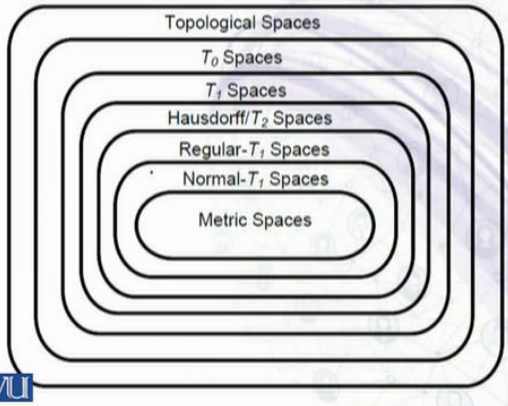
## Topology

$T_i$ Properties: An Overview

## $T_i$ Properties: An Overview

An overview of $T_i$ properties is given in this topic through a diagram. This diagram illustrates the relationship between the $T_i$ spaces we studied in previous topics.

## $T_i$ Properties: An Overview




## Topology

Urysohn's Lemma

## Urysohn's Lemma

In this topic, one of the most classical and important results of topology will be discussed. This result is named as "Urysohn's Lemma".
Let us state this lemma.

## Urysohn's Lemma

**Urysohn's Lemma:** Let $X$ be a normal space; $A$ and $B$ be two disjoint closed subsets of $X$. Let $[a, b]$ be a closed interval in $\mathbb{R}$. Then there exists a continuous map

$$f : X \longrightarrow [a, b]$$

such that $f(x) = a \ \forall \ x \in A$ and $f(x) = b \ \forall \ x \in B$.

**Urysohn's Lemma: (Strong Form)**
$X$ is a normal space if and only if for every two disjoint closed subsets $A$ and $B$ of $X$. There exists a continuous map

$$f : X \longrightarrow [a, b]$$

such that $f(x) = a \ \forall \ x \in A$ and $f(x) = b \ \forall \ x \in B$.

## Urysohn's Lemma

One important application of Urysohn's Lemma is the partial solution of metrization problem. Here we have this solution in the form of "Urysohn's Metrization Theorem".

**Urysohn's Metrization Theorem:**
Every normal-$T_1$ space with countable basis is metrizable.

Another version of Urysohn's Metrization Theorem is

**Urysohn's Metrization Theorem:**
Every regular-$T_1$ space with countable basis is metrizable.

## Topology

Compact Spaces

## Compact Spaces

Recall the definitions of cover, subcover, and open cover.

**Cover:**
Let $(X, \mathcal{T})$ be a topological space. Let $\mathcal{C} = \{C_i\}$ be a class of subsets of $X$, such that

$$X = \bigcup_{i \in I} C_i,$$

then $\mathcal{C}$ is called a cover of $X$.

## Compact Spaces

**Subcover:**
A subclass $\mathcal{S}$ of a cover $\mathcal{C}$ of $X$ which is also a cover is called a subcover.

**Open cover:**
A cover $\mathcal{C}$ of $X$ is said to be an open cover if and only if $\mathcal{C} \subset \mathcal{T}$.

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## Compact Spaces

**Compact Space:**
A topological space $(X, \mathcal{T})$ is said to be compact if every open cover $\mathcal{C}$ of $X$ contains a finite subcover $\mathcal{S}$ of $X$.

## Topology

Examples: Compact Spaces

## Examples: Compact Spaces

**Example 1:**
A set $X$ with indiscrete topology $\mathcal{T}$ is compact.

**Reason:**
Since there is only one open cover of $X$ with respect to indiscrete topology and that is

$$\{X\}$$

which is it self finite.

Dr.

## Examples: Compact Spaces

**Example 2:**
A set $X$ with topology $\mathcal{T}$ containing finite number subsets of $X$ is compact.

**Reason:**
Since every open cover $\mathcal{C}$ of $X$ is a subclass of $\mathcal{T}$ and $\mathcal{T}$ itself finite so $\mathcal{C}$ is finite too and any subcover $\mathcal{S}$ of $\mathcal{C}$ is also finite.

## Examples: Compact Spaces

**Example 3:**
A finite set $X$ with any topology $\mathcal{T}$ is compact.

**Reason:**
Since every topology of a finite set $X$ is finite and from "Example 2" $X$ is compact.

## Examples: Compact Spaces

**Example 4:**
An infinite set $X$ with discrete topology $\mathcal{T}$ is not compact.
**Reason:**

$\mathcal{T}_{dis} = \mathcal{P}(X)$

$\mathcal{C} = \{\{x\} \mid x \in X\} \cup \{\{x_1, x_2\}\}$ (fin.)

$X = \bigcup_{i \in I} C_i$ $\quad C_i = \{x_i\}$

## Topology

Open Intervals in $\mathbb{R}$

## Open Intervals in $\mathbb{R}$

In this topic we will discuss the compactness of open intervals in $\mathbb{R}$ with respect to the usual topology.

**Fact:**
An open interval in $\mathbb{R}$ with respect to usual topology is not compact.

An immediate consequence of this fact is

$(\mathbb{R}, \mathcal{T}_u)$ is not compact.

## Open Intervals in $\mathbb{R}$

**Proof:**
Consider $(\mathbb{R}, \mathcal{T}_u)$ and $(a, b)$ be an open interval in $\mathbb{R}$.
Consider the following covering of $(a, b)$.

$$\mathcal{C} = \left\{\left(a + \frac{b-a}{n+2}, a + \frac{b-a}{n}\right)\right\}$$

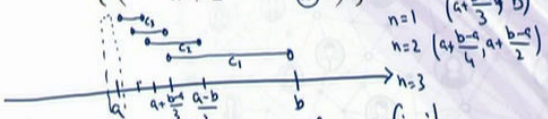
## Open Intervals in $\mathbb{R}$

**Proof:**
Consider $(\mathbb{R}, \mathcal{T}_u)$ and $(a, b)$ be an open interval in $\mathbb{R}$.
Consider the following covering of $(a, b)$.

$$\mathcal{C} = \left\{\left(a + \frac{b-a}{n+2}, a + \frac{b-a}{n}\right)\right\} \quad n \in \mathbb{Z}^+$$

$n=1 \quad (a + \frac{b-a}{3}, b)$

$n=2 \quad (a + \frac{b-a}{4}, a + \frac{b-a}{2})$




$\nexists$ subcover of $\mathcal{C}$ that is finite

So $(a, b)$ is not compact.

## Topology

Cofinite Topology and Compactness

## Cofinite Topology and Compactness

In this topic we will study the compactness of cofinite space, a set $X$ with cofinite topology $\mathcal{T}_{cof}$.

**Fact:**
A set $X$ with cofinite topology $\mathcal{T}_{cof}$ is compact.

## Cofinite Topology and Compactness

**Proof:** If $X$ is finite. Then $\mathcal{T}_{cof} = \mathcal{P}(X)$

$\Rightarrow (X, \mathcal{T}_{cof})$ is compact.

But if $X$ is not finite. Then

Consider an open cover $\mathcal{U}$ of $X$.

Now let $\emptyset \neq O \in \mathcal{U}$ $\because$ $\mathcal{U}$ is an open cover

$\Rightarrow O$ is open. $\Rightarrow O^c$ is finite

Now for every $x \in O^c$ choose an element of $\mathcal{U}$ $O_x$ containing $x$.

$\{O_x \mid x \in O^c\}$ is finite

## Cofinite Topology and Compactness

$\because$ $D^c$ is finite.

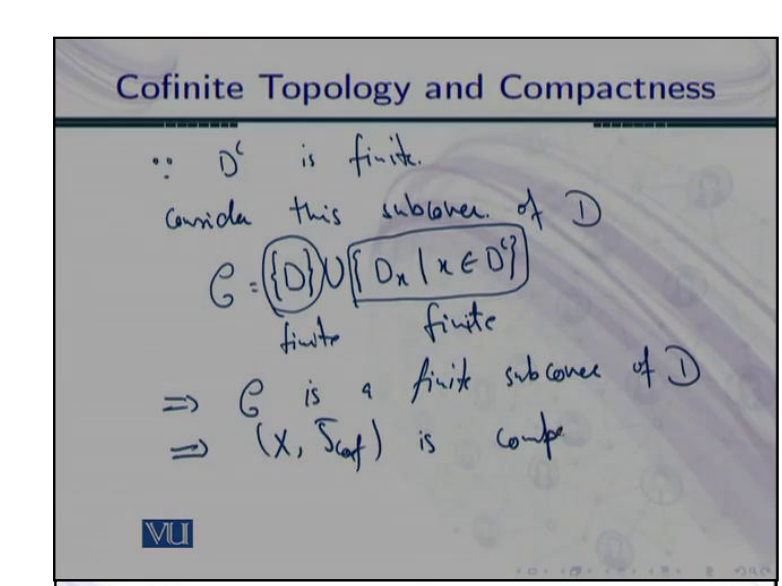
Consider this subcover of $\mathcal{D}$

$\mathcal{C} = \{D\} \cup \{D_x \mid x \in D^c\}$

finite, finite

$\Rightarrow$ $\mathcal{C}$ is a finite subcover of $\mathcal{D}$

$\Rightarrow$ $(X, \mathcal{T}_{cof})$ is compact



## Topology

Compactness and Open Collection

## Compactness and Open Collection

The aim of this topic is to show that compactness of some space depends on the collection of open subsets of the space.

Consider $\mathbb{R}$, now

- $(\mathbb{R}, \mathcal{T}_u)$ is not compact.
- $(\mathbb{R}, \mathcal{T}_{indis})$ is compact.
- $(\mathbb{R}, \mathcal{T}_{dis})$ is not compact.
- $(\mathbb{R}, \mathcal{T}_{cof})$ is compact.

## Topology

Theorem: Closed Subspace of Compact Space

## Theorem: Closed Subspace...

**Theorem:**
Every closed subspace of a compact space is compact.

**Proof:**
Let $(X, \mathcal{T})$ be a compact space and $A$ be a closed subset of $X$.
So $(A, \mathcal{T}_A)$ be closed subspace of $X$, where $\mathcal{T}_A$ is subspace topology on $A$.
Consider an open cover $\mathcal{D}$ of $A$, i.e.

## Theorem: Closed Subspace...

$$A = \bigcup_{i \in I} D_i$$

where $D_i \in \mathcal{D} \subset \mathcal{T}_A$.
Now for each $D_i$ there exists $U_i \in \mathcal{T}$ such that $D_i = A \cap U_i$.
So

$$A = \bigcup_{i \in I} D_i = \bigcup_{i \in I} (A \cap U_i)$$

But $A$ is closed in $X$ so $A^c$ is an open subset of $X$. And we have an open covering of $X$.

$$X = A^c \cup (\bigcup_{i \in I} U_i)$$

## Theorem: Closed Subspace...

$$A = \bigcup_{i \in I} D_i$$

where $D_i \in \mathcal{D} \subset \mathcal{T}_A$.
Now for each $D_i$ there exists $U_i \in \mathcal{T}$ such that $D_i = A \cap U_i$.
So

$$A = \bigcup_{i \in I} D_i = \bigcup_{i \in I} (A \cap U_i) \Rightarrow A \subset \bigcup_{i \in I} U_i$$

But $A$ is closed in $X$ so $A^c$ is an open subset of $X$. And we have an open covering of $X$.

$$X = A^c \cup (\bigcup_{i \in I} U_i)$$

## Theorem: Closed Subspace...

But $X$ is compact. So every open cover of $X$ has a finite subcover, i.e. there exists $J \underset{\text{finite}}{\subset} I$ such that

$$X = A^c \cup (\bigcup_{j \in J} U_j)$$

$\Longrightarrow$

$$A \subset \bigcup_{j \in J} U_j$$

$A \cap U_j = D_j$

$\Longleftrightarrow$

$$A = A \cap (\bigcup_{j \in J} U_j) = \bigcup_{j \in J} D_j$$

Proved.

## Theorem: Closed Subspace...

But $X$ is compact. So every open cover of $X$ has a finite subcover, i.e. there exists $J \underset{\text{finite}}{\subset} I$ such that

$$X = A^c \cup (\bigcup_{j\in J} U_j)$$

$$\Longrightarrow$$

$$A \subset \bigcup_{j\in J} U_j$$

$$\Longleftrightarrow$$

$$A = A \cap (\bigcup_{j\in J} U_j) = \bigcup_{j\in J} D_j \quad (A \cap U_j = D_j)$$

A is compact.

Proved.

## Topology

Compact Subspace of Hausdorff Space

## Compact Subspace of Hausdorff...



**Theorem:**
Let $(A, \mathcal{T}_A)$ be compact subspace of a Hausdorff space $(X, \mathcal{T})$.
Then $A$ is closed in $X$.
i.e.
Every compact subspace of a Hausdorff space is closed.

## Compact Subspace of Hausdorff...

**Proof:** Let $(X, \mathcal{T})$ is Hausdorff.
Consider $(A, \mathcal{T}_A)$ a subspace of $(X, \mathcal{T})$
Compact.
Now we have to show that A is closed in X

$\Longleftrightarrow$ $A = \bar{A}$

$\Longleftrightarrow$ $\nexists\ x \in A^c = X - A$ s.t $x \in \bar{A}$

$\longrightarrow \Longleftrightarrow$ for every $x \in A^c = X - A$ $\exists\ U_x$ (open subset of X containing x) s.t $U_x \cap A = \emptyset$

## Compact Subspace of Hausdorff...

Let $x \in A^c$
Given $(X, \mathcal{T})$ is Hausdorff
So for every $y \in A \subset X$
$\exists$ $U_{xy}$, $U_y$ open subsets of X containing x and y resp.
s.t $U_{xy} \cap U_y = \emptyset$
Now for every $y \in A$ such disjoint pair of subsets exists. one containing x and other containing y

## Compact Subspace of Hausdorff...

So we have an open cover of A

$$A \subset \bigcup_{y\in A} U_y$$

$$\Rightarrow A = \bigcup_{y\in A} (A \cap U_y)$$

But given A is compact. Then this cover has a finite subcover.
$\exists\ y_1, y_2, \ldots, y_n \in A$ s.t

$$A = \bigcup_{i=1}^{n} (A \cap U_{y_i})$$

## Compact Subspace of Hausdorff...

$$\Rightarrow A \subset \bigcup_{i=1}^{n} U_{y_i} = V \text{ (Say)}$$

Now $U_x = \bigcap_{i=1}^{n} U_{xy_i}$ an open subset of X containing x.

More than this one $U_x \cap V = \emptyset$

$$\Rightarrow U_x \cap A = \emptyset$$

$\Rightarrow$ A is closed.

## Topology

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Compact Space under Continuous Map

## Compact Space under Cont. Map

**Theorem:**
Let $(X, \mathcal{T})$ be a compact space and

$$f : (X, \mathcal{T}) \to (Y, \mathcal{T}')$$

be a continuous map. Then $f(X)$ is compact.
i.e.
Image of compact space under a continuous map is compact.

## Compact Space under Cont. Map

**Proof:** Let $f : (X, \mathcal{T}) \xrightarrow[\text{cont.}]{\text{compact}} (Y, \mathcal{T}')$

Now $f(X) \subset Y$

Now consider an open cover of $f(X)$

$f(X) \subset \bigcup_{i \in I} U_i \qquad U_i \in \mathcal{T}'$

$\because$ $f$ is cont. So $\forall\ U_i \in \mathcal{T}'$

$f^{-1}(U_i) \in \mathcal{T}$ Now $D_i = f^{-1}(U_i)$

## Compact Space under Cont. Map

$X = \bigcup_{i \in I} f^{-1}(U_i) = \bigcup_{i \in I} D_i$

So we have an open cover of $X$

But given, $X$ is compact.

So $\exists$ a finite subcover of this cover

i.e. $\exists\ J \subset_{\text{finite}} I$ s.t

$X = \bigcup_{j \in J} D_j = \bigcup_{j \in J} f^{-1}(U_j)$

$\Rightarrow$ $\boxed{f(X) \subset \bigcup_{j \in J} U_j}$ $\Rightarrow$ $f(X)$ is compact.

## Topology

Compactness and Homeomorphism

## Compactness and Homeomorphism

**Corollary I:**
Let $(X, \mathcal{T})$ be a compact space and $(Y, \mathcal{T}')$ be a Hausdorff space. Then a continuous map

$$f : (X, \mathcal{T}) \to (Y, \mathcal{T}')$$

is a closed map.

## Compactness and Homeomorphism

**Proof:**
Let $A$ be a closed subset of $X$. It is given, $X$ is compact and we know that a closed subspace of a compact space is compact.

So $A$ is compact. Now $f$ is continuous and we know that image of a compact space under a continuous map is compact so $f(A)$ is a compact subspace of $Y$.

Now $Y$ is Hausdorff and is has been proved that every compact subspace of a Hausdorff is closed, so $f(A)$ is closed.

So $f$ is a closed map.

## Compactness and Homeomorphism

**Corollary II:**
Let $(X, \mathcal{T})$ be a compact space and $(Y, \mathcal{T}')$ be a Hausdorff space. Then a bijective continuous map

$$f : (X, \mathcal{T}) \to (Y, \mathcal{T}')$$

is a homeomorphism.

## Topology

Application: Compactness and Homeomorphism

## Application: Compactness and...

**Exercise I:**
Consider $(\mathbb{R}, \mathcal{T}_u)$. Show that there exists no homeomorphism between an open interval of $\mathbb{R}$ and a closed interval of $\mathbb{R}$.

**Exercise II:**
Prove or disprove.

$$[0;1) \approx S^1$$

## Application: Compactness and...

Ex(1) We know that the open intervals in $(\mathbb{R}, \mathcal{T}_u)$ are not compact.
Moreover the only compact subsets of $\mathbb{R}$ are closed and bounded subsets.
Compact $[a,b]$ $(c,d)$ Not compact.
if $\exists$ $f: [a,b] \xrightarrow[\text{cont}]{\text{bijection}} (c,d)$
Then $f([a,b]) = (c,d)$
compact Not compact
$\not\exists$ $f$:

## Application: Compactness and...

Ex(2) $[0,1) \not\approx S^1$
$\because$ $S^1$ is compact and $[0,1)$ is not compact.

$f: [0,1) \xrightarrow[\text{cont}]{\text{bij}} S^1$ ✓
$x \longmapsto (\cos 2\pi x, \sin 2\pi x)$
Can we say $\Rightarrow$ $f$ is homeomorphism No
$\because$ $[0,1)$ is not compact.

## Topology

Finite Union of Compact Spaces

## Topology

Finite Union of Compact Spaces

## Finite Union of Compact Spaces

**Theorem:**
Let $(X, \mathcal{T})$ be a topological space. And

$$\{(X_i, \mathcal{T}_i) | i \in I\}$$

be a finite family of compact subspaces of $X$.
Then

$$\bigcup_{i \in I} X_i$$

is compact.

## Finite Union of Compact Spaces

**Proof:** Let $X_1, X_2 \subset_{\text{compact}} X$
Now $X_1 \cup X_2$ we want to show that this union is compact.
Let us consider an open cover of $X_1 \cup X_2$. $X_1 \cup X_2 \subset \bigcup_{i \in I} U_i$ $U_i \in \mathcal{T}$
$\Rightarrow$ $X_1 \subset \bigcup_{i \in I} U_i$ and $X_2 \subset \bigcup_{i \in I} U_i$
But $X_1, X_2$ both compact subspaces of $X$.

## Finite Union of Compact Spaces

$\Rightarrow$ $X_1 \subset \bigcup_{j \in J} U_j$ and $X_2 \subset \bigcup_{j' \in J'} U_{j'}$
where $J, J' \subset I$ finite
$\Rightarrow$ $X_1 \cup X_2 \subset \left(\bigcup_{j \in J} U_j\right) \cup \left(\bigcup_{j' \in J'} U_{j'}\right)$
$\Rightarrow$ $X_1 \cup X_2$ is compact.

## Topology

Compact Hausdorff Space is Normal

## Compact Hausdorff Space is Normal

**Theorem:**
Let $(X, \mathcal{T})$ be a compact Hausdorff space. Then $(X, \mathcal{T})$ is normal.
i.e.
For every pair of disjoint closed subsets $F_1, F_2$ of a compact Hausdorff space $X$ there exists a pair of disjoint open subsets $V_{F_1}, V_{F_2}$ of $X$ containing $F_1, F_2$ respectively.

## Compact Hausdorff Space is Normal

**Proof:**
Let

$$F_1, F_2 \underset{\text{closed}}{\subset} X$$

such that

$$F_1 \cap F_2 = \emptyset$$

Consider an open covering of $F_1$

$$F_1 \subset \bigcup_{x \in F_1} V_x$$

Now $F_1$ is a closed subset of compact space $X$, so $F_1$ is compact.

## Compact Hausdorff Space is Normal

So there exist $x_1, x_2, \ldots, x_n \in F_1$ such that

$$F_1 \subset \bigcup_{i=1}^{n} V_{x_i} = V_{F_1}(\text{say})$$

Given, $X$ is Hausdorff, so for an element $y$ of $F_2$ there exists $U_{y_{x_i}}$, an open set containing $y$ such that

$$U_{y_{x_i}} \cap V_{x_i} = \emptyset \quad .$$

So $\bigcap_{i=1}^{n} U_{y_{x_i}}$ is an open set containing $y$ and disjoint from $V$.

## Compact Hausdorff Space is Normal

Continue with this construction one can find an open covering of $F_2$ by taking the union of such open subsets over $y \in F_2$, i.e.

$$F_2 \subset \bigcup_{y \in F_2} \left( \bigcap_{i=1}^{n} U_{y_{x_i}} \right)$$

Again $F_2$ is compact so this cover can be reduced to a finite subcover, i.e. there exist

$$y_1, y_2, \ldots, y_m \in F_2$$

## Compact Hausdorff Space is Normal

such that

$$F_2 \subset \bigcup_{j=1}^{m} \left( \bigcap_{i=1}^{n} U_{y_{j_{x_i}}} \right) = U_{F_2}(\text{say})$$

And we have

$$V_{F_1} \bigcap U_{F_2} = \emptyset$$

$\Longrightarrow$
$X$ is normal.

## Topology

Separated Sets

## Separated Sets

**Def I:**
Let $A$ and $B$ be two subsets of a topological space $(X, \mathcal{T})$. Then $A$ and $B$ are said to be separated sets if and only if

$$A \cap B = \emptyset$$
and
$$A' \cap B = \emptyset$$
and
$$A \cap B' = \emptyset$$

## Separated Sets

**Def II:**
Let $A$ and $B$ be two subsets of a topological space $(X, \mathcal{T})$. Then $A$ and $B$ are said to be separated sets if and only if

$$\overline{A} \cap B = \emptyset$$
and
$$A \cap \overline{B} = \emptyset$$

## Separated Sets

**Def III:**
Let $A$ and $B$ be two subsets of a topological space $(X, \mathcal{T})$. Then $A$ and $B$ are said to be separated sets if and only if there exist open subsets $U_A$ and $U_B$ of $X$ containing $A$ and $B$ respectively such that

$$A \cap U_B = \emptyset$$
and
$$B \cap U_A = \emptyset$$

## Topology

Examples: Separated Sets

## Examples: Separated Sets

**Example 1:**
Consider $(\mathbb{R}, \mathcal{T}_u)$. Then $A = (0, 1)$ and $B = (4, 9]$ are separated sets.
**Reason:** (By Definition I)
Since

$$(0, 1) \cap (4, 9] = \emptyset$$

and

$$(0, 1)' \cap (4, 9] = \emptyset$$

and

$$(0, 1) \cap (4, 9]' = \emptyset$$

## Examples: Separated Sets

**Example 1:**
Consider $(\mathbb{R}, \mathcal{T}_u)$. Then $A = (0, 1)$ and $B = (4, 9]$ are separated sets.
**Reason:** (By Definition I)
Since

$$(0, 1) \cap (4, 9] = \emptyset$$

and

$$(0, 1)' \cap (4, 9] = \emptyset$$

and

$$(0, 1) \cap (4, 9]' = \emptyset$$

## Examples: Separated Sets

Let discuss "Example 1" with "Definition III" of separated sets.
There exist $U_A = (-0.5, 2)$ and $U_B = (3, 10)$ two open subsets of $\mathbb{R}$ containing $A = (0, 1)$ and $B = (4, 9]$ respectively such that

$$A \cap U_B = \emptyset$$
and
$$B \cap U_A = \emptyset$$

So $A$ and $B$ are separated sets of $(\mathbb{R}, \mathcal{T}_u)$.

## Examples: Separated Sets

**Example 2:**
Consider $(\mathbb{R}, \mathcal{T}_u)$. Then $A = (-1, 3]$ and $B = [3, 5)$ are not separated sets.
**Reason:**
By "Definition I" both sets should be disjoint but this is not the case here, i.e.

$$(-1, 3] \cap [3, 5) = \{3\} \neq \emptyset$$

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## Examples: Separated Sets

**Example 3:**
Consider $(\mathbb{R}, \mathcal{T}_u)$. Then $A = (-1, 3)$ and $B = [3, 5)$ are not separated sets.
**Reason:** (By Definition I)
Both sets are disjoint, i.e.

$$(-1, 3) \cap [3, 5) = \emptyset$$

But at the same time both sets should not contain each others limit points. But here

$$(-1, 3)' \cap [3, 5) = \{3\}$$

## Examples: Separated Sets

**Example 3:**
Consider $(\mathbb{R}, \mathcal{T}_u)$. Then $A = (-1, 3)$ and $B = [3, 5)$ are not separated sets.
**Reason:** (By Definition I)
Both sets are disjoint, i.e.

$$(-1, 3) \cap [3, 5) = \emptyset$$

But at the same time both sets should not contain each others limit points. But here

$$(-1, 3)' \cap [3, 5) = \{3\} \neq \emptyset$$

## Topology

Connected Spaces

## Connected Spaces

**Connected Set:**
A subset $A$ of a topological space $(X, \mathcal{T})$ is said to be connected iff there exists no pair of nonempty open subsets $U$ and $V$ of $X$ such that

$$A \cap U \quad \text{and} \quad A \cap V$$

are nonempty disjoint sets and

$$A = (A \cap U) \cup (A \cap V)$$

## Connected Spaces

**Disconnected Set:**
A subset $A$ of a topological space $(X, \mathcal{T})$ is said to be disconnected if it is not connected, i.e.
if there exists a pair of nonempty open subsets $U$ and $V$ of $X$ such that $A \cap U$ and $A \cap V$ are nonempty disjoint sets and

$$A = (A \cap U) \cup (A \cap V).$$

## Connected Spaces

**Connected Space:**
A topological space $(X, \mathcal{T})$ is said to be connected if it can not be written as union of two nonempty open disjoint subsets $U$ and $V$ of $X$.

## Connected Spaces

**Disconnected Space:**
A topological space $(X, \mathcal{T})$ is said to be disconnected if it can be written as union of two nonempty open disjoint subsets $U$ and $V$ of $X$.

## Connected Spaces

**Connected Space:** (Def. II)
A topological space $(X, \mathcal{T})$ is said to be connected if

$$X = D_1 \cup D_2$$

such that $D_1$, $D_2$ both open and

$$D_1 \cap D_2 = \emptyset$$

implies that either

$$D_1 = \emptyset \quad \text{or} \quad D_2 = \emptyset.$$

## Topology

Examples: Connected Spaces

## Examples: Connected Spaces

**Example 1:** Consider $(\mathbb{R}, \mathcal{T}_u)$. Then $A = (0,1) \cup (3,5]$ is disconnected subset of $\mathbb{R}$.
**Reason:**
Since there exists a pair of nonempty open subsets

$$U = (0,1) \text{ and } V = (3,6)$$

of $\mathbb{R}$ such that
$A \cap U$ and $A \cap V$ are nonempty disjoint sets and

$$A = (A \cap U) \cup (A \cap V).$$

## Examples: Connected Spaces

**Example 1:** Consider $(\mathbb{R}, \mathcal{T}_u)$. Then $A = (0,1) \cup (3,5]$ is disconnected subset of $\mathbb{R}$.
**Reason:**
Since there exists a pair of nonempty open subsets

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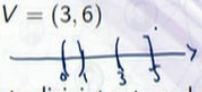


of $\mathbb{R}$ such that
$A \cap U$ and $A \cap V$ are nonempty disjoint sets and

$$A = (A \cap U) \cup (A \cap V).$$

$(0,1) \cup (3,5]$

## Examples: Connected Spaces

**Example 2:** Consider $X = \{a, b, c, d, e\}$ with
$\mathcal{T} = \{\emptyset, \{a\}, \{a, b\}, \{a, b, c\}, \{a, d, e\}, \{a, b, d, e\}, X\}$.
Then $A = \{c, e\}$ is disconnected subset of $X$.
**Reason:**
Since there exists a pair of nonempty open subsets
$U = \{a, b, d, e\}$ and $V = \{a, b, c\}$ of $X$ such that

$$A \cap U = \{e\} \quad \text{and} \quad A \cap V = \{c\}$$

are nonempty disjoint sets and

$$A = (A \cap U) \cup (A \cap V) = \{e\} \cup \{c\}.$$

## Examples: Connected Spaces

**Example 2:** Consider $X = \{a, b, c, d, e\}$ with
$\mathcal{T} = \{\emptyset, \{a\}, \{a, b\}, \{a, b, c\}, \{a, d, e\}, \{a, b, d, e\}, X\}$.
Then $A = \{c, e\}$ is disconnected subset of $X$.
**Reason:**
Since there exists a pair of nonempty open subsets
$U = \{a, b, d, e\}$ and $V = \{a, b, c\}$ of $X$ such that

$$A \cap U = \{e\} \quad \text{and} \quad A \cap V = \{c\}$$

$\{e\} \cap \{c\} = \emptyset$

are nonempty disjoint sets and

$$A = (A \cap U) \cup (A \cap V) = \{e\} \cup \{c\} = \{c, e\}$$

## Examples: Connected Spaces

**Example 3:** Consider $X = \{a, b, c, d, e\}$ with
$\mathcal{T} = \{\emptyset, \{a, b\}, \{a, b, c\}, X\}$. Then $X$ is a connected space.
**Reason:**
Since there exists no pair of nonempty open disjoint subsets

$$U \text{ and } V$$

of $X$ such that

$$X = U \cup V.$$



## Examples: Connected Spaces

**Example 4:** Consider $X = \{a, b, c, d, e\}$ with
$\mathcal{T} = \{\emptyset, \{a, b\}, \{d, e\}, \{a, b, c\}, \{a, b, d, e\}, X\}$. Then $X$ is a disconnected space.
**Reason:**
Since there exist nonempty open disjoint subsets

$$U = \{a, b, c\} \text{ and } V = \{d, e\}$$

of $X$ such that

$$X = U \cup V.$$



## Topology

Connected Spaces and Open Closed Sets



## Connected Spaces and Open...

Recall
**Connected Space:**
A topological space $(X, \mathcal{T})$ is said to be connected iff

$$X = D_1 \cup D_2$$

such that $D_1$, $D_2$ both open and disjoint implies that either

$$D_1 = \emptyset \text{ or } D_2 = \emptyset.$$

## Connected Spaces and Open...

**Disconnected Space:**
A topological space $(X, \mathcal{T})$ is said to be disconnected iff

$$X = D_1 \cup D_2$$

such that $D_1$, $D_2$ both open and disjoint implies that neither $D_1$ is empty nor $D_2$ is empty.

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## Connected Spaces and Open...

$X = D_1 \cup D_2$ $\quad D_1 \cap D_2 = \emptyset$

disconnected $\quad$ Both are open

$D_1 \neq \emptyset$ $\quad D_2 \neq \emptyset$

$D_1$ is open

$\Rightarrow D_1^c$ is closed.

$\quad\quad\quad

## Connected Set and its Closure

Say $A \cap N = \emptyset$

$\Rightarrow A \subset A \cap M \Rightarrow A \subset M$

$\Rightarrow \bar{A} \subset \bar{M}$

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Now $A \subset B \subset \bar{A} \subset \bar{M}$

But $\bar{M} \cap N = \emptyset$

So $B \cap N = \emptyset \Rightarrow (N = \emptyset)$

$\Rightarrow$ B is connected. And this implies that $\bar{A}$ is connected.



## Topology

Image of Connected Space

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## Image of Connected Space

**Theorem:**
Let $f$ be a continuous function from $(X, T)$ to $(Y, T')$ and $A$ be a connected subset of $(X, T)$. Then $f(A)$ is connected.
i.e.

Image of a connected space under a continuous map is connected.



## Image of Connected Space

Proof $f: (X, T) \xrightarrow[\text{Cont.}]{} (Y, T')$

$A \subset X$ Connected

Now we have to show that $f(A)$ is connected.

Let $f(A)$ is disconnected

$\Rightarrow \exists$ a separation of $f(A)$

So $f(A) = B \cup C$ B, C both Nonempty + open: Disjoint



## Image of Connected Space

Now $A \subset f^{-1}(B \cup C) = f^{-1}(B) \cup f^{-1}(C)$

$\Rightarrow A = A \cap (f^{-1}(B) \cup f^{-1}(C))$

$= (A \cap f^{-1}(B)) \cup (A \cap f^{-1}(C))$

open open open open

$\because B \neq \emptyset \quad C \neq \emptyset \quad f(A) = B \cup C$

$\Rightarrow A \cap f^{-1}(B) \neq \emptyset$ and $A \cap f^{-1}(C) \neq \emptyset$

$\Rightarrow (A \cap f^{-1}(B)) \cap (A \cap f^{-1}(C)) = \emptyset \quad B \cap C = \emptyset$



## Image of Connected Space

So we have a separation of A.

But A is connected.

$\exists$ no separation

$\Rightarrow f(A)$ is connected $\because \exists$ no separation of $f(A)$



## Topology

Application: Image of Connected Set



## Application: Image of...

**Exercise I:**
Consider $(\mathbb{R}, \mathcal{T}_u)$ and show that there exists no homeomorphism between $(0, 2)$ and $(0, 1] \cup [3, 4)$.

**Exercise II:**
Prove or disprove

$$(0, 1) \approx (0, 1]$$

## Application: Image of...

Ex ① Let $f: (0,2) \longrightarrow (0,1] \cup [3,4)$

bijective + bicont.

$\because$ $f$ is cont and w.r.t $\mathcal{T}_u$ on $\mathbb{R}$

$A=(0,2)$ is connected. $\Rightarrow$ $f(A)$ is connected.

But $(0,1] \cup [3,4)$ is not connected.

$f(A) \longrightarrow (0,1]$ or $[3,4)$

$\Rightarrow \not\exists\ f \Rightarrow (0,2) \not\simeq (0,1] \cup [3,4)$



## Application: Image of...

Ex ② $(0,1) \overset{??}{\simeq} (0,1]$

Ans. $\exists$ no such $f$ bijective + bicont.

Let $\exists\ f: (0,1] \longrightarrow (0,1)$ biject + bicont

$1 \in (0,1] \Rightarrow f(1) \in (0,1)$

$f': (0,1] \setminus \{1\} \longrightarrow (0,1) \setminus \{f(1)\} = (0, f(1)) \cup (f(1), 0)$

connected; biject + bicont (Not possible); disconnected

$\Rightarrow (0,1) \not\simeq (0,1]$



## Topology

Connected Subsets of Real Line



## Connected Subsets of Real Line

**Theorem:**
A subset $A$ of $(\mathbb{R}, \mathcal{T}_u)$ is connected if and only if $A$ is an interval.

**Proof:**
Let $A$ be a nonempty connected subset of $\mathbb{R}$.
If $A$ contains only one point then we have nothing to prove.
Let $A$ contains two points, say

$$a, b \in A$$



## Connected Subsets of Real Line

Consider $a < b$.
If every real number $c$, between $a$ and $b$, belongs to $A$.
Then $A$ is an interval. Proved.
If this is not the case, i.e. there exists some $c \in (a, b)$ such that

$$c \notin A$$

Then we can write $A$ as

$$A = (A \cap (-\infty, c)) \cup (A \cap (c, \infty))$$

This provides us a separation of $A$, i.e. a nonempty, open, disjoint pair of sets whose union is $A$.



## Connected Subsets of Real Line

This implies that $A$ is disconnected and it is contradiction of the fact that $A$ is connected.
So for all $c \in (a, b)$, $\Longrightarrow$ $c \in A \Longrightarrow$ $A$ is an interval.
**Converse:**
Let $A$ be an interval. i.e.

$$A = [a, b]$$

Now let $A$ is disconnected. This means there exists a separation of $A$, i.e. there exist $B$, $C$ nonempty, open, disjoint subsets such that

$$A = B \cup C$$



## Connected Subsets of Real Line

Now let

$$d = \sup\{c \in A | [a, c] \subset B\} \in A$$

Now one thing is very much clear here that $d \neq b$.
Let $d \in B$, now since $B$ is open so there exists an open set $U_d$ containing $d$ such that

$$U_d \subset B.$$

So we can find a real number $d_1 \in U_d$ greater than $d$ such that

$$[a, d_1] \subset B$$



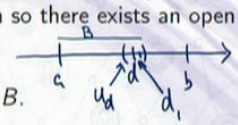
## Connected Subsets of Real Line

Now let

$$d = \sup\{c \in A | [a, c] \subset B\} \in A$$

($M$)

Now one thing is very much clear here that $d \neq b$.
Let $d \in B$, now since $B$ is open so there exists an open set $U_d$ containing $d$ such that




$U_d \subset B$.

So we can find a real number $d_1 \in U_d$ greater than $d$ such that

$[a, d_1] \subset B \Rightarrow d_1 \in M$

But $d$ is sup $M$ and $d_1 > d$

## Connected Subsets of Real Line

But this is a contradiction of the fact the $d$ is the supremum. So $d \notin B$ and it means

$$d \in C$$

Again $C$ is open so there exists an open set $V_d$ containing $d$ such that $V_d \subset C$.
This implies that there exists a real number $d_2 \in V_d$, less than $d$. So

$$[a, d_2] \subset B \quad \Rightarrow \quad d_2 \in B \cdot$$

And we have again a contradiction of the fact $B \cap C = \emptyset$.
So $A$ is connected.

## Topology

**Connectedness and Fixed Point Theorem**

## Connectedness and Fixed Point...

**Fixed Point:**
A point $x \in X$ is said to be a fixed point of a map $f : X \to X$ iff $f(x) = x$.

**Example:**

1. $f : \mathbb{R} \to \mathbb{R}$, defined as $f(x) = x^2$. Here 0 is a fixed point of $f$. 1 is also a fixed point of $f$.
2. $g : \mathbb{R} \to \mathbb{R}$, defined as

$$g(x) = x + 1$$

$g$ has no fixed point.

## Connectedness and Fixed Point...

**Theorem:**
Consider $\mathbb{R}$ with usual topology and

$$I = [0, 1]$$

be closed interval in $\mathbb{R}$. Then every continuous map

$$f : I \to I$$

has at least one fixed point.

## Connectedness and Fixed Point...

**Geometrical Interpretation:**

$f: I \xrightarrow[\text{Cont}]{} I \qquad \exists\ x \in I$ s.t $f(x) = x$

## Connectedness and Fixed Point...

**Proof:** $f: I \xrightarrow[\text{Cont.}]{} I$

If $f(0) = 0$ or $f(1) = 1$ then, proved.

Now suppose that $f(0) > 0$ and $f(1) < 1$

Now the graph of $f$ is connected.

i.e $G = \{(x, f(x)) \in \mathbb{R}^2 \mid 0 \le x \le 1,\ 0 \le f(x) \le 1\} \subset \mathbb{R}^2$ Connected.

Now consider $M, N \subset G$ defined as:

## Connectedness and Fixed Point...

$\to M = \{(x, f(x)) \mid x < f(x)\} \subset G$

$M \neq \emptyset \quad \because (0, f(0)) \in M$ since $f(0) > 0$

$\to N = \{(x, f(x)) \mid x > f(x)\} \subset G$

$N \neq \emptyset \quad (1, f(1)) \in N$ since $f(1) < 1$

Both are open subsets.

$\to M \cap N = \emptyset$

## Connectedness and Fixed Point...

[Suppose $\exists$ no fixed point of $f$. i.e $\nexists\ x \in I$ s.t $x = f(x)$]

So $G = M \cup N$ Separation of $G$.

$\Rightarrow G$ is disconnected. ✗

But $G$ is connected.

So $\exists\ x \in I$ s.t $f(x) = x$.

## Topology

Union of Connected Subsets

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## Union of Connected Subsets

**Theorem:**
Let $A$ and $B$ be two connected subsets of a topological spaces $(X, T)$ such that

$$x \in A \cap B$$

than

$$C = A \cup B$$

is a connected subset of $X$.

## Union of Connected Subsets

**Proof:** $A, B \subset X$ connected s.t $x \in A \cap B$

Now $C = A \cup B \leftarrow$ is connected !!

Let $C$ is not connected. i.e

$\exists$ a separation of $C$ say $C = M \cup N$ s.t $M, N$ both
- open
- Nonempty.
- disjoint.

Now $x \in C$ either $x \in M$ or $x \in N$

## Union of Connected Subsets

Say $x \in M$ Then

$A \subset M$ also $B \subset M$

$\because x \in A$ $\because x \in B$

$\Rightarrow A \cup B \subset M$

$\Rightarrow C \subset M \Rightarrow N = \emptyset$

So $\nexists$ no such separation of $C$.

So $C$ is connected

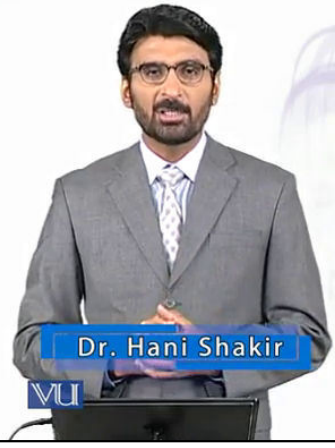
## Union of Connected Subsets

**Theorem:**
The union of a collection of connected subspaces of $(X, T)$, that having a point common, is connected.

## Topology

Connected Component

## Connected Component




**Def:**
Consider a topological space $(X, T)$. Let $a \in X$ and $C$ be a connected subset of $X$ containing $a$.
Then

$$C_a = \bigcup_{a \in C} C$$

is called the connected component of $X$ containing $a$.

## Connected Component

(4) A space $(X, \mathfrak{I})$ is connected iff it has only one connected component.

**Remarks:**

1. The connected component $C_a$ is the largest connected subset of $X$ containing $a$.
2. If $b \in C_a$ then $C_b = C_a$.
3. $X$ can be written as disjoint union of its connected components. i.e.

$$X = \coprod_{C_i} C_i$$

## Topology

Examples: Connected Component

## Examples: Connected Component

**Example 1:** Consider $X = \{a, b, c, d, e\}$ with

$$\mathcal{T} = \{\emptyset, \{a, b\}, \{d, e\}, \{a, b, c\}, \{a, b, d, e\}, X\}$$

$C_a = \bigcup_{a \in C} C$ $\{a\}, \{a,b\}, \{a,c\}, \{a,b,c\}$

$\{a,b,d\} = (B \cap \{a,b,c\}) \cup (B \cap \{d,e\})$

$B$

$= \{a,b,c\}$

$\Rightarrow C_a = C_b = C_c$

Now $C_d = \bigcup_{d \in C} C$ $\{d\}$ $\{d,e\}$ $\{c,d,e\}$ Not connected

$= \{d,e\} = C_e$ $X = C_a \sqcup C_d$.

## Examples: Connected Component

**Example 2:** Consider $X = \{a, b, c, d\}$ with

$$\mathcal{T} = \{\emptyset, \{a, b\}, \{a, b, c\}, X\}$$

$C_a = \bigcup_{a \in C} C = X = C_b = C_c = C_d$

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## Topology

Connected Component is Closed

## Connected Component is Closed

**Theorem:**
Let $C$ be a connected component of $X$ then

$$C = \overline{C}$$

i.e.
A Connected component is a closed subset of $X$.

## Connected Component is Closed

**Proof:** Let $C$ be a connected component of $X$.

$C \subset \overline{C}$ —①

But we know that $C$ is connected and if a set is connected then its closure is connected too.

$\Rightarrow \overline{C}$ is connected.

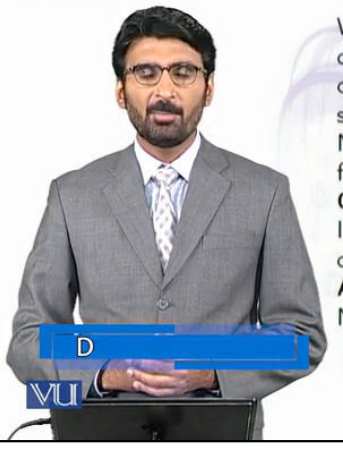
But $C$ is the max connected subset.

$\Rightarrow C \subset \overline{C}$ —② $\Rightarrow C = \overline{C}$ □

## Topology

Connected Component is not Open

## Connected Component is not Open



We have discussed about the closed nature of connected component of a topological space $(X, \mathcal{T})$.
Now one may ask the following question.
**Question:**
Is a connected component $C$ of $X$ open?
**Answer:**
No, not in general.

## Connected Component is not Open

**Example:** Let $\mathbb{Q} \subset \mathbb{R}$ $(\mathbb{R}, \mathcal{T}_u)$

Now the connected components of $(\mathbb{Q}, \mathcal{T}_{\mathbb{R}_u})$ are single tons

$\mathbb{Q} = \coprod_{x \in \mathbb{Q}} \{x\}$

Now $\{x\} \subset \mathbb{Q}$ Not open

$\therefore U_x \subset_{\text{open}} \mathbb{Q}$ s.t $U_x \subset \{x\}$

## Topology

Locally Connected Spaces



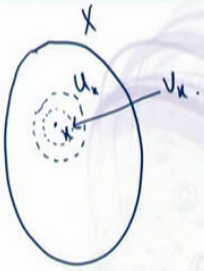
## Locally Connected Spaces

**Def:**
Let $(X, \mathcal{T})$ be a topological space. $X$ is said to be locally connected at a point $x \in X$ iff for every $U_x$, open subset of $X$ containing $x$, there exists $V_x$, an open subset of $X$ containing $x$, such that

- $V_x \subset U_x$
- $V_x$ is connected.



## Locally Connected Spaces






## Locally Connected Spaces

**Def:**
A topological space $(X, \mathcal{T})$ locally connected iff it is locally connected at each of its points.
i.e.
For all $x \in X$ and for every $U_x$, open subset of $X$ containing $x$, there exists $V_x$, an open subset of $X$ containing $x$, such that

- $V_x \subset U_x$
- $V_x$ is connected.



## Topology

Examples: Locally Connected Spaces



## Examples: Locally Connected...

**Example 1:** Consider $X = \{a, b, c, d\}$ with

$$\mathcal{T} = \{\emptyset, \{a, b\}, \{a, b, c\}, X\}$$



## Examples: Locally Connected...

**Example 1:** Consider $X = \{a, b, c, d\}$ with

$$\mathcal{T} = \{\emptyset, \{a, b\}, \{a, b, c\}, X\}$$

"a" $U_a = \{a, b, c\}$ $U_a = \{a, b\}$ connected.

"b" $U_b$

## Examples: Locally Connected...

**Example 2:** A discrete space is locally connected.

$X$ $\quad \mathcal{T}_{dis} = \mathcal{P}(X)$

$x \in X \quad U_x \in \mathcal{P}(X)$

$V_x = \{x\} \subset U_x.$

## Topology

Connectedness and Locally Connectedness

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## Connectedness and Locally...

**Questions:**

1. Does locally connectedness imply connectedness?
2. Does connectedness imply locally connectedness?

Dr. H



## Connectedness and Locally...

Locally Connectedness $\not\Longrightarrow$ Connectedness

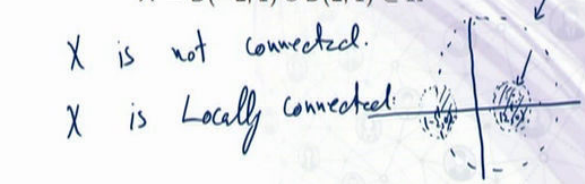
**Example:**

$X = B(-2;1) \cup B(2;1) \subset \mathbb{R}^2$

$X$ is not connected.

$X$ is Locally Connected.





## Connectedness and Locally...

Connectedness $\not\Longrightarrow$ locally Connectedness

**Example:**

$$Y = \{(x,0) | x \in [0,1]\} \bigcup \left[ \bigcup_{n \in \mathbb{Z}^+} \left\{ \left(x, \frac{x}{n}\right) | x \in [0,1] \right\} \right] \subset \mathbb{R}^2$$

$n=1 \quad \{(x,x) \mid x \in [0,1]\}$

$n=2 \quad \{(x, \frac{x}{2}) | x \in [0,1]\}$

Now $Y$ is connected.

## Connectedness and Locally...

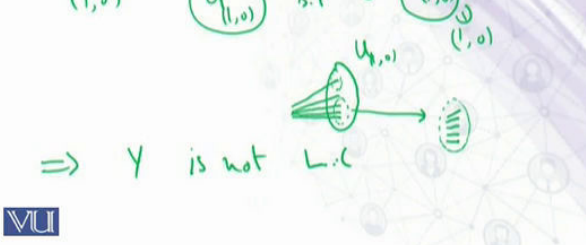
Now $Y$ is not Locally Connected.

$(1,0) \in Y$ s.t $Y$ is not L.C at $(1,0)$

$(1,0)$ $U_{(1,0)}$ s.t $\nexists$ $V_{(1,0)}$ Not connected $\cup$ $(1,0)$

$U_{(1,0)}$

$\Rightarrow$ $Y$ is not L.C



## Topology

Path Connected Spaces



## Path Connected Spaces

**Path:**
Let $(X, \mathcal{T})$ be a topological space and $x, y \in X$. A path from $x$ to $y$ in $X$ is a continuous map

$$p : [0,1] \to X$$

such that

$$p(0) = x$$

and

$$p(1) = y.$$

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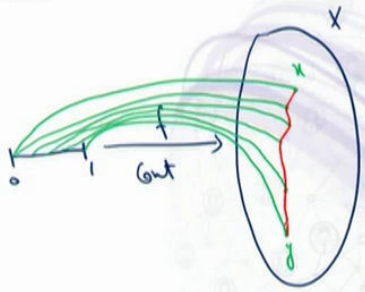
## Path Connected Spaces




## Path Connected Spaces

**Path Connected Space:**
A topological space $(X, \mathcal{T})$ is said to be path connected iff for every $x, y \in X$ there exists a path $p$ from $x$ to $y$ in $X$.

**Examples:**

- $(\mathbb{R}, \mathcal{T}_u)$
- $(\mathbb{R}^2, \mathcal{T}_u)$

## Topology

Facts: Path Connected Spaces: 1

## Facts: Path Connected Spaces: 1

**Theorem:**
Let $(X, \mathcal{T})$ be a path connected topological space then $(X, \mathcal{T})$ is connected.

i.e.

Path connectedness implies connectedness.

Dr. Hani

## Facts: Path Connected Spaces: 1

Proof Let $x \in X$

$\because$ $X$ is path connected. So $\forall\, y \in X$

$\exists$ a path from $x$ to $y$.

i.e. $\gamma_y : I \to X$ s.t $\gamma_y(0) = x$, $\gamma_y(1) = y$

($I = [0,1]$)

So $X = \bigcup_{y \in X} \gamma_y(I)$

Now $\gamma_y(0) = x \quad \forall\, y \in X$

$\Rightarrow x \in \gamma_y(I) \quad \forall\, y \in X.$

## Facts: Path Connected Spaces: 1

$\Rightarrow x \in \bigcap_{y \in X} \gamma_y(I)$

$\Rightarrow X = \bigcup_{y \in X} \gamma_y(I)$

$\to$ is connected.

$\gamma_y(I)$ is connected

$\because$ $I$ is connected

And we know that The image of a connected space is connected under a cont map.

## Facts: Path Connected Spaces: 1

Remarks ① Connectedness $\not\Rightarrow$ Path connectedness

② $D \subsetneq_{\text{open}} \mathbb{R}^n$

if $D$ is connected $\Rightarrow D$ is p

## Topology

Facts: Path Connected Spaces: 2

## Facts: Path Connected Spaces: 2

**Proposition:**
Let $(X, \mathcal{T})$ be a path connected topological space and

$$f : (X, \mathcal{T}) \longrightarrow (Y, \mathcal{T}')$$

be a continuous map. Then $f(X)$ is path connected.

## Facts: Path Connected Spaces: 2

Proof: $f : (X, \mathcal{T}) \xrightarrow[\text{Cont}]{} (Y, \mathcal{T}')$

$\downarrow$ Path connected.

$f(X)$ is path connected ??

Let $a, b \in f(X) \Rightarrow \exists\ x, y \in X$ s.t $f(x) = a$, $f(y) = b$

Now $\because$ $X$ is path connected.

So $\exists$ $p : I \xrightarrow[\text{Cont}]{} X$ $\quad p(0) = x$, $p(1) = y$

## Facts: Path Connected Spaces: 2

$q = f \circ p : I \longrightarrow f(X)$ ✓

$q(0) = f \circ p(0) = f(p(0)) = f(x) = a$ ✓

$q(1) = f \circ p(1) = f(p(1)) = f(y) = b$ ✓

$\Rightarrow$ $f(X)$ is path connected.

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